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Collaborative threat intelligence: Enhancing IoT security through blockchain and machine learning integration

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ABSTRACT

Ensuring robust security in the Internet of Things (IoT) landscape is of paramount importance. This research article presents a novel approach to enhance IoT security by leveraging collaborative threat intelligence and integrating blockchain technology with machine learning (ML) models. The iOS application acts as a central control centre, facilitating the reporting and sharing of detected threats. The shared threat data is securely stored on a blockchain network, enabling ML models to access and learn from a diverse range of threat scenarios. The research focuses on implementing Random Forest, Decision Tree classifier, Ensemble, LSTM, and CNN models on the IoT23 dataset within the context of a Collaborative Threat Intelligence Framework for IoT Security. Through an iterative process, the models' accuracy is improved by reducing false negatives through the collaborative threat intelligence system. The article investigates the implementation details, privacy considerations, and the seamless integration of ML-based techniques for continuous model improvement. Experimental evaluations on the IoT23 dataset demonstrate the effectiveness of the proposed system in enhancing IoT security and mitigating potential threats. The research contributes to the advancement of collaborative threat intelligence and blockchain technology in the context of IoT security, paving the way for more secure and reliable IoT deployments.

1. Introduction

The security of Internet of Things (IoT) systems has become a critical concern due to the widespread adoption of interconnected devices across various domains. Traditional security measures often fall short in protecting against the dynamic and evolving threat landscape faced by IoT networks (Ammar et al., 2018). To address this challenge, innovative approaches leveraging advanced technologies such as ML algorithms, and blockchain applications have gained significant attention (Durani et al., 2018; Minoli and Occhiogrosso, 2018; Dorri et al., 2017, 2019). In this study, we propose a novel solution that combines the power of ML algorithms, an iOS application acting as a control center, and blockchain technology to enhance the security of IoT systems. We utilized the IoT23 dataset for experimental purposes (Garcia et al., 2020). This dataset provides a comprehensive and representative collection of IoT-related data, enabling us to train and evaluate ML models specifically tailored to IoT security challenges. By pre-processing and transforming the IoT23 dataset, we ensure that the data is in a suitable format for training the ML models. The core idea behind our research is to leverage the collaborative capabilities of the iOS application and blockchain technology to enhance threat detection and mitigation. When a threat is detected or suspicious data is identified, an autogenerated push notification is sent to the control centre, which is the iOS application. This push notification contains vital information related to the suspicious data or device, enabling immediate action to be taken. Leveraging Apple's push notification

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service, real-time notifications ensure that security experts or administrators are promptly alerted to potential threats. Upon receiving the push notification, the control centre initiates the necessary actions to verify the threat's authenticity. The data associated with the suspicious device or event is carefully examined to determine its validity. If the threat is confirmed, the relevant device information and threat data are securely transmitted back to the blockchain using smart contracts. This collaborative approach not only improves the accuracy of the ML models but also gradually reduces false negatives (FN). By leveraging the collective intelligence and continuous feedback loop facilitated by the collaboration between the iOS application, ML models, and blockchain, we can maximize the accuracy of threat detection while minimizing false negatives. This collaborative threat intelligence framework allows the ML models to learn from a broader range of real-world threat scenarios and adapt to emerging threats effectively. The experimental evaluation of our framework is conducted using the IoT23 dataset, and the results demonstrate the significant impact of collaborative learning and blockchain integration in enhancing IoT security. By continuously updating and retraining the ML models with the verified threat data stored on the blockchain, we achieve higher accuracy rates and minimize the occurrence of false negatives. Our research presents a comprehensive framework that combines ML algorithms, an iOS application acting as a control centre, and blockchain technology to enhance the security of IoT systems. By fostering collaboration and leveraging the power of blockchain-based threat intelligence, we can significantly improve the accuracy of threat detection while reducing false negatives. The experimental results and analysis demonstrate the effectiveness of our approach and highlight the potential of collaborative learning and blockchain integration in addressing the challenges of IoT security.

Our research makes several significant contributions to the field of IoT security. Firstly, we propose a novel collaborative threat intelligence framework that combines the power of ML algorithms, an iOS application acting as a control center, and blockchain technology. This integration allows for real-time threat detection, reporting, and response, enhancing the overall security posture of IoT systems. Secondly, we demonstrate the effectiveness of our approach through extensive experimentation and evaluation using the IoT23 dataset. The results highlight the substantial impact of collaborative learning and blockchain integration in improving the accuracy of threat detection while reducing false negatives. Our research findings provide valuable insights into the potential of collaborative threat intelligence for enhancing IoT security.

The subsequent sections of this paper are methodically structured to facilitate a comprehensive understanding of our research. Section 2 delves into foundational concepts and offers a backdrop of the prevailing research landscape. Section 3 critically reviews the extant literature in domains of IoT security, collaborative threat intelligence, and the confluence of blockchain with ML paradigms. In Section 4, we articulate the intricacies of our proposed framework, while Section 5 elucidates the methodological approach, encompassing the integration mechanisms of the framework's pivotal components. Additionally, this section demystifies the experimental design, dataset employed, and the evaluative metrics adopted. Section 6 is devoted to the presentation and interpretative analysis of our empirical findings. We reflect upon the overarching implications and insights in Section 7, drawing the paper to a close. Lastly, Section 8 charts out prospective avenues for future exploration and enhancement in this research domain.

2. Essential concepts and research background

Internet of Things (IoT) has emerged as a transformative paradigm, enabling seamless connectivity and communication among a vast array of physical devices, ranging from sensors and actuators to everyday objects (Pradhan and Singh, 2021). This interconnected ecosystem has revolutionized various domains, including healthcare, transportation, manufacturing, and smart homes. However, the rapid proliferation of IoT devices has also brought forth significant security challenges (Siddhartha et al., 2020; Ahsan et al., 2023). Securing IoT systems is crucial due to the sensitive nature of the data and the potential risks associated with unauthorized access or malicious attacks. Traditional security approaches often struggle to address the unique characteristics of IoT, such as resource constraints, heterogeneity, and the scale of devices and data (Ambarkar and Shekokar, 2020). Consequently, new strategies and technologies are needed to ensure the robustness and resilience of IoT security measures (Hussain et al., 2020). ML algorithms have demonstrated great potential in enhancing IoT security (Huynh et al., 2015; Nazir et al., 2020; Wajahat et al., 2023). By analyzing large volumes of data generated by IoT devices, ML models can identify patterns, anomalies, and potential threats. ML techniques, such as random forest, decision trees, LSTM, CNN, and RNN, have been applied to detect intrusion, classify malicious behavior, and predict security incidents in IoT networks (Muneer et al., 2023; Martins et al., 2022). The integration of ML algorithms into IoT security frameworks can provide proactive threat detection, real-time monitoring, and adaptive defense mechanisms (Sarker, 2022; Kaushik, 2023; Huynh et al., 2015). Blockchain technology has gained significant attention in the context of IoT security (Haque et al., 2023). As a distributed and immutable ledger, blockchain offers several key advantages for securing IoT systems. It provides transparency, traceability, and tamper resistance, making it suitable for maintaining a decentralized record of IoT device interactions, access control policies, and security-related events (Namane and Ben Dhaou, 2022; Padmavathy et al., 2022). Blockchain enables secure data sharing, integrity verification, and decentralized consensus mechanisms, thereby reducing the reliance on central authorities and enhancing the overall security and trustworthiness of IoT ecosystems (Ghosh et al., 2023; Nawaz et al.). iOS operating system powering Apple's mobile devices plays a crucial role in enhancing IoT security. With its robust security features, strict app vetting process, and hardware-level protections, iOS provides a secure foundation for developing IoT applications (Howell et al., 2022; Huynh et al., 2015). iOS devices can act as control centers, enabling users to monitor, manage, and interact with IoT devices and data securely. The iOS ecosystem offers secure communication channels, such as Apple Push Notification service (APNs), which can be leveraged for real-time threat alerts and secure data transmission between IoT devices and the control centre (Dehalwar et al., 2022; Huang et al., 2022). By combining ML algorithms, blockchain technology, and iOS applications, we propose a collaborative threat intelligence framework to enhance IoT security. This integrated approach leverages ML for proactive threat detection, blockchain for secure data sharing and storage, and iOS as a user-friendly control center. The framework fosters collaboration among users, enables real-time threat response, and enhances the overall security posture of IoT systems.

3. Literature review

The security of Internet of Things (IoT) systems has been an active area of research due to the increasing concerns over the vulnerabilities and threats faced by interconnected devices. In this section, we review the existing literature and highlight the key contributions and limitations of prior studies related to IoT security, ML algorithms, blockchain technology, and iOS applications. Several research efforts have focused on applying ML algorithms for enhancing IoT security. For instance, Ravi et al. (2022) proposed a deep learning-based approach to detect anomalies in IoT network traffic, achieving high accuracy in identifying malicious activities (Ravi et al., 2022). Similarly, authors utilized ML techniques to predict and prevent IoT device vulnerabilities by analyzing their behavior and identifying potential attack patterns (Ali et al., 2022; Pan et al., 2022). While these studies demonstrate the effectiveness of ML in detecting threats, they often rely on centralized architectures and lack the collaborative aspect necessary for comprehensive threat intelligence. Blockchain technology

has emerged as a promising solution for secure and decentralized data management (Latif et al., 2021). In recent years, there has been a growing interest in leveraging ML algorithms for IoT security. Bagaa et al. (2020) proposed a hybrid ML approach that combines supervised and unsupervised learning techniques to detect anomalies and intrusions in IoT networks (Bagaa et al., 2020). Their study demonstrated improved detection rates and reduced false positive rates compared to traditional rule-based methods. Similarly, Idrissi et al. (2021) explored the use of deep learning algorithms, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), for detecting IoT malware and botnet attacks (Idrissi et al., 2021). Their results showed promising accuracy and robustness in identifying malicious activities. These works highlight the potential of ML algorithms in detecting threats in IoT environments, providing valuable insights for our research.

In the context of IoT security, various studies have explored the application of blockchain to enhance trust, integrity, and privacy. Alzubi et al. (2021) proposed a blockchain-based framework to securely authenticate IoT devices and ensure data integrity in a distributed environment (Alzubi, 2021). Another study by Al-Omrani et al. (2023) leveraged blockchain to create a tamper-proof audit trail for IoT data, enabling transparent and verifiable data exchanges (Al-Omrani and Humayun, 2023). However, these works primarily focus on data integrity and authentication aspects of IoT security, and do not fully exploit the potential of collaborative threat intelligence. Blockchain technology has also been extensively investigated in the context of IoT security. Shi et al. (2021) proposed a blockchain-based framework that enables secure and auditable data sharing among IoT devices, ensuring data integrity and privacy (Shi et al., 2021). Their study emphasized the importance of decentralized consensus mechanisms and tamper-resistant data storage for enhancing trust in IoT ecosystems. Additionally, El Bekkali et al. (2023) presented a blockchain-based architecture for IoT systems that enables secure device authentication and data provenance (El Bekkali et al., 2023). Their framework addressed the challenges of scalability and interoperability in IoT deployments. These studies contribute to the understanding of blockchain's potential in securing IoT systems and inform the design of our collaborative threat intelligence framework.

The role of iOS applications in enhancing IoT security has also been investigated. Authors developed an iOS application that enables users to monitor and control devices remotely, enhancing user accessibility and device management (Kucerik, 2023; Nazir et al., 2018). Similarly, the Authors proposed iOS applications with a user-friendly interface for real-time threat reporting and incident response in IoT environments (Dragonas et al., 2023; Hutchinson et al., 2023; Kamaruzaman and Rashid, 2023). These studies demonstrate the value of iOS applications in improving user experience and facilitating interaction with IoT systems. However, they do not explore the integration of iOS applications with ML algorithms and blockchain technology for collaborative threat intelligence. While individual studies have contributed to the understanding of different aspects of IoT security, collaborative threat intelligence frameworks that integrate ML algorithms, blockchain technology, and iOS applications remain relatively unexplored. Our research aims to fill this gap by proposing a comprehensive framework that leverages the strengths of these technologies to enhance IoT security. By combining the power of ML algorithms in threat detection, the decentralized and secure nature of blockchain for threat data storage, and the user-friendly interface of an iOS application, our framework enables collaborative threat intelligence and addresses the limitations of existing approaches. Previous research has made significant contributions to various aspects of IoT security, ML algorithms, blockchain technology, and iOS applications. However, there is a lack of comprehensive frameworks that integrate these technologies for collaborative threat intelligence in IoT systems as shown in Table 1. Our research builds upon these prior works and presents a novel framework that combines ML algorithms, an iOS application acting as a control centre, and blockchain technology to enhance the security of IoT systems. By fostering collaboration and leveraging the power of blockchain-based threat intelligence, our framework aims to address the challenges and limitations of existing approaches and contribute to the advancement of IoT security research.

Table 2 provides a comprehensive overview of key studies in the field of collaborative threat intelligence for enhanced IoT security, emphasizing the contributions, methodologies, and main findings of each. The first study, Saxena and Gayathri's work (Saxena and Gayathri, 2022), focuses on cyber threat intelligence (CTI) challenges and introduces the Collaborative Cyber Threat Information Exchange (CCTI) system. This collaborative framework, relying on artificial intelligence computing on the blockchain, proposes an effective means to address challenges in CTI by facilitating the secure exchange of security-related information across various industries.

The second study by Sarhan et al. (2022) delves into the security posture of IoT ecosystems, proposing the Hierarchical Blockchain-Based Federated Learning (HBFL) framework for collaborative IoT intrusion detection. The authors underscore the importance of data privacy, secure architectures, and scalable deployments. Through hierarchical federated learning on the blockchain, the HBFL framework ensures secure and privacy-preserving intrusion detection in collaborative IoT environments, as evidenced by its successful implementation and evaluation using a key IoT dataset.

Arshad et al.'s work (Arshad et al., 2019), the third study in the table, introduces the Collaborative Intrusion Detection Framework (COLIDE) for IoT. The study addresses the vulnerability of resource-constrained sensor devices in IoT to misuse by intruders. Through collaboration among sensor and border nodes, COLIDE achieves efficient intrusion detection in IoT systems. The implementation and experimentation using Contiki OS demonstrated its effectiveness, particularly in mitigating energy and processing overheads within IoT systems.

The fourth study by Sarhan et al. (2023) proposes a federated learning-based cyber threat intelligence sharing scheme for network intrusion detection. This study addresses the challenge of designing machine learning-based detection systems using heterogeneous data from various sources and organizations. The scheme utilizes federated learning and a common data format to allow multiple organizations to collaborate without sharing sensitive user information. The results demonstrate the efficiency of the proposed framework in designing a universal machine learning model for classifying various traffic types without the need for inter-organizational data exchange.

The fifth study, presented by Kumar et al. (2022), introduces the P2TIF framework – a combination of scalable blockchain and deep learning modules – for privacy-preserved threat intelligence in Industrial IoT. This study tackles security, privacy, and scalability issues in IIoT, offering a scalable blockchain module for secure IIoT data communication and employing deep learning techniques for threat detection. The proposed P2TIF framework is validated using two different network data sources, demonstrating high efficiency and scalability.

In addition to these five studies, Table 2 includes our study, which introduces a novel Collaborative Threat Intelligence Framework integrating blockchain, machine learning, and an iOS application. This framework addresses the identified research gap by providing a holistic synthesis of collaborative threat intelligence for IoT security. By combining human intelligence, blockchain-secured data sharing, and machine learning analysis, our study contributes to advancing the state of the art in collaborative threat intelligence.

4. CTIF-IoT: Collaborative threat intelligence framework for IoT security

We outline a thorough architecture in this section that is designed to improve the security of Internet of Things (IoT) ecosystems. With the essential role of human expertise, our method makes use of a synergistic

Table 1
Overview of research initiatives in IoT security: A focus on ML, blockchain, and iOS applications.

Study	Blockchain	Machine Learning (ML)	iOS application
Pan et al. (2022)		✓	
Ravi et al. (2022)			✓
Ali et al. (2022)		✓	
Latif et al. (2021)	✓		
Bagaa et al. (2020)		✓	
Idrissi et al. (2021)		✓	
Alzubi (2021)	✓		
Al-Omrani and Humayun (2023)	✓		
Shi et al. (2021)	✓		
El Bekkali et al. (2023)	✓	✓	
Kucerik (2023), Dragonas et al. (2023), Hutchinson et al. (2023), Kamaruzaman and Rashid (2023)			✓
This study	✓	✓	✓

Table 2
Comparative overview of relevant literature in collaborative threat intelligence for enhanced IoT security.

Study	Key contributions	Methodology/Framework	Main findings
Saxena and Gayathri (2022)	Discussion of CTI challenges. Proposal of CCTI system.	Artificial intelligence computing on blockchain.	Addressed challenges in CTI. Proposed a collaborative system for threat information exchange.
Sarhan et al. (2022)	Proposal of HBFL framework for collaborative IoT intrusion detection.	Hierarchical federated learning on blockchain.	Ensured secure and privacy-preserving intrusion detection in collaborative IoT environments. Demonstrated feasibility with implementation and evaluation using a key IoT dataset.
Arshad et al. (2019)	Introduction of COLIDE framework for collaborative IoT intrusion detection.	Collaboration among resource-constrained sensor and border nodes.	Efficient intrusion detection in IoT systems with collaboration. Implementation and experimentation using Contiki OS demonstrated effectiveness with respect to energy and processing overheads.
Sarhan et al. (2023)	Proposal of a federated learning-based cyber threat intelligence sharing scheme.	Utilization of federated learning and a common data format.	Efficient design of ML-based network intrusion detection system without inter-organizational data exchange. Evaluation using NetFlow datasets showcased effective classification of various traffic types.
Kumar et al. (2022)	Introduction of P2TIF framework for privacy-preserved threat intelligence.	Combination of scalable blockchain and deep learning modules.	Mitigated security, privacy, and scalability issues in IIoT. Validation using ToN-IoT and IoT-Botnet datasets revealed high efficiency and scalability.
Our study	Comprehensive Collaborative Threat Intelligence Framework integrating blockchain, machine learning, and iOS application.	Integration of human intelligence, blockchain-secured data sharing, and machine learning analysis.	Fills the research gap by providing a holistic synthesis of collaborative threat intelligence for IoT security.

combination of ML, an iOS control centre, and blockchain technology. While continuously learning and developing to deal with novel safety concerns, the framework operates in real-time, enabling quick threat detection and response. Key components of CTIF-IoT are shown in Fig. 1. The framework begins with the deployment of IoT devices, which form the interconnected network of devices in various domains. These devices generate data and face potential security threats due to their connected nature. The first key component of the framework is the ML models. These models are trained using relevant datasets, such as the IoT23 dataset, to learn patterns, identify threats, and classify security incidents in real-time.

The second component is the iOS application acting as a control centre. The application provides a user-friendly interface for threat reporting and interaction with the ML models. It enables users to report detected threats, providing essential information such as threat type, timestamp, and source. Additionally, the application leverages features like push notifications and secure communication channels to ensure timely threat detection and response. The third crucial component is the integration of blockchain technology. When a threat is detected and classified by the ML models, the information is shared between the ML models and the iOS application. If the ML models determine that a threat is present, the threat data, along with device information, is securely transmitted to the blockchain network using smart contracts. The blockchain network acts as a decentralized and immutable repository for threat data, ensuring its integrity and facilitating collaboration among different participants.

The CTIF-IoT emphasizes the collaborative nature of threat intelligence. The shared threat data stored on the blockchain network serves as a collective repository that continuously enhances the ML models' accuracy and effectiveness. The ML models can learn from a wider range of threat scenarios, adapt to emerging threats, and address false negatives by incorporating the shared intelligence. The iterative process of the CTIF-IoT involves periodic updates and retraining of the ML models using the shared threat data from the blockchain network. This allows the models to stay up to date with evolving threat landscapes and improve their detection capabilities over time.The CTIF-IoT establishes a collaborative ecosystem where IoT devices, ML models, the iOS application, and blockchain technology work together to enhance IoT security. It enables real-time threat detection, efficient threat reporting, secure data transmission, and continuous model improvement, ultimately contributing to the overall protection of IoT systems.

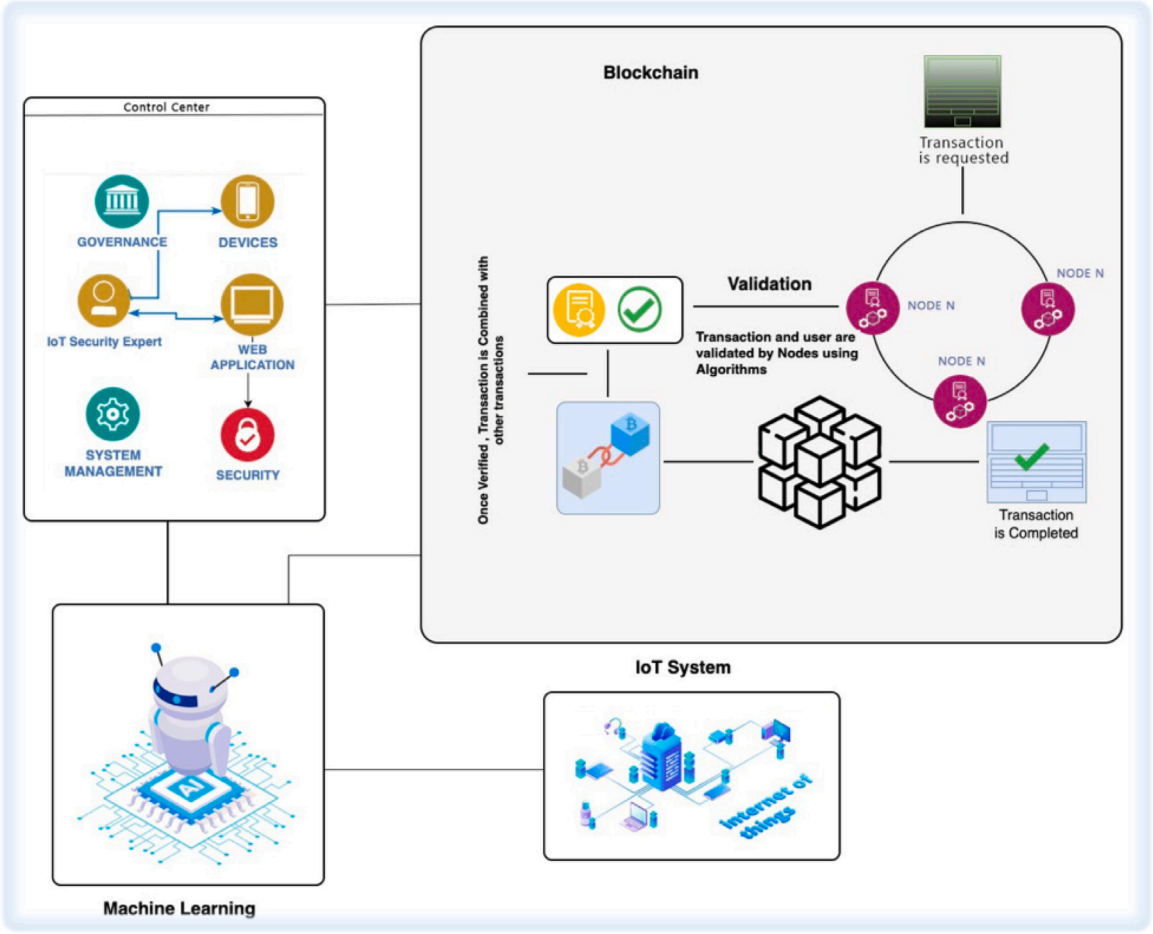


Fig. 1. Conceptual framework of proposed framework.

4.1. Main components of framework

4.1.1. IoT Security Ecosystem

The IoT Security Ecosystem serves as the foundation of our framework, representing the vast array of interconnected devices and systems that constitute the Internet of Things. This includes everything from smart home devices and sensors to industrial machinery and autonomous vehicles. Within this complex network, the primary concern revolves around ensuring the security and integrity of data and operations. Threats can vary widely, from unauthorized access to data breaches, making it essential to have a robust security framework in place.

In our study, we delve into the intricate layers of the IoT security ecosystem, as illustrated comprehensively in Fig. 2. At the outset, the User Layer emerges as the primary interface, consisting of end-users, devices, and their digital environs. Our innovative approach introduces Blockchain for its inherent transparency and indelibility, fortifying the authentication and authorization processes against potential vulnerabilities like unauthorized device interactions or inadvertent misconfigurations. Coupled with this, ML offers an insightful grasp on user patterns, detecting and responding to aberrations in real-time.

Diving deeper, the Proximity Network serves as a nexus between user devices and expansive networks, facilitating short-range communications like Bluetooth and Wi-Fi. This bridge, under our research's aegis, harnesses Blockchain to validate information exchanges meticulously and ascertain their authenticity. ML supplements this by discerning patterns and forestalling potential security lapses. The Public Network, akin to a vast global web, necessitates rigorous encryption and trusted communication channels, for which Blockchain proves instrumental. The Provider Cloud—central repositories for IoT data, demands stringent security protocols. Here, Blockchain ensures data transparency and resistance to tampering, while ML consistently surveils access patterns, alerting of any discrepancies.Blockchain guarantees device legitimacy, while ML ensures system integrity by incessantly learning from and rectifying any network inconsistencies.

4.1.2. Machine learning layer

Situated at the heart CTIF-IoT, the ML Layer is responsible for continuously monitoring the data traffic within the IoT Security Ecosystem. Leveraging a diverse set of ML algorithms, this layer examines patterns and anomalies in real-time. Its role is to identify potential security threats based on deviations from established data patterns. ML acts as the first line of defense in threat detection.

4.1.3. iOS control center

Serving as a vital bridge between the ML Layer and human expertise, the iOS Control Center functions as the system's communication hub. When the ML Layer detects a potential security threat, it immediately triggers an alert in the form of an Apple push notification. This notification is sent to the designated IoT security expert researcher, ensuring that security incidents are promptly brought to the attention of human experts. As shown in Table 3.

4.1.4. Blockchain technology

The Blockchain Technology component introduces transparency, immutability, and security into the framework. It operates as a decentralized ledger that records validated security incidents in a tamper-proof manner. This ledger is a critical asset for accumulating expert

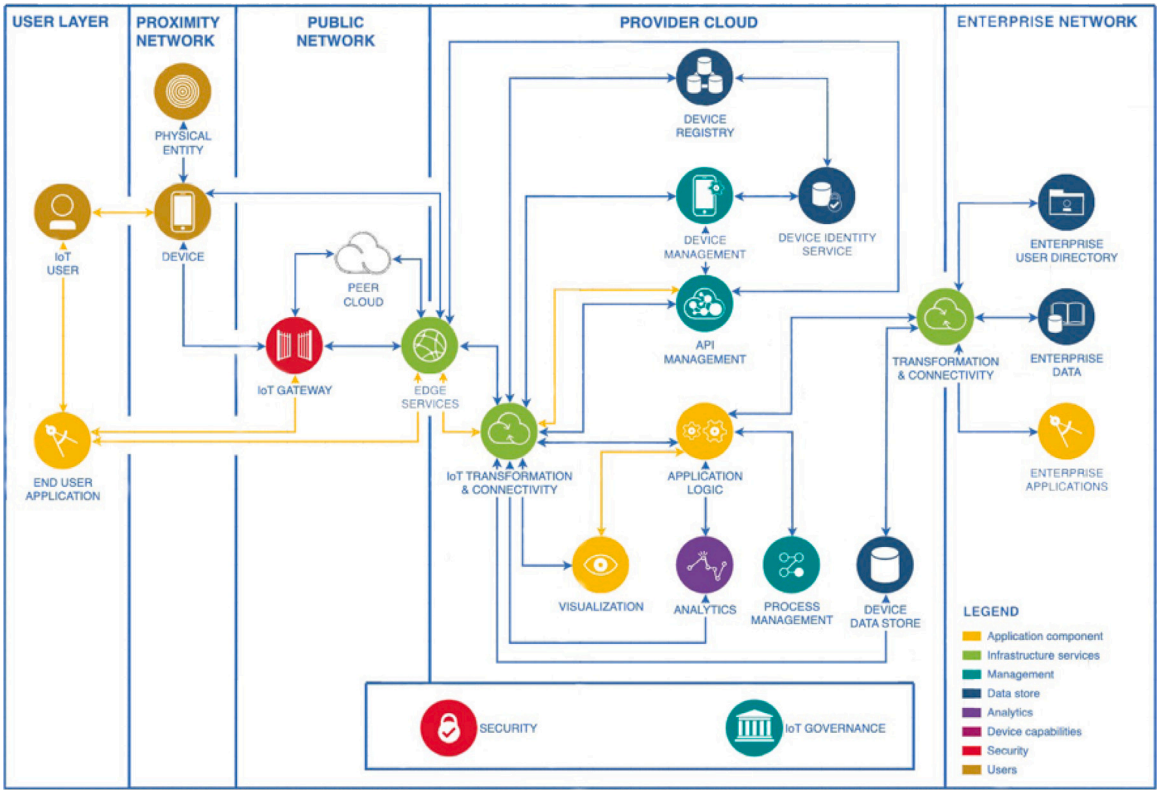


Fig. 2. IoT ecosystem.

Table 3
Role of iOS app in the framework and advantages.

Role of iOS app in the framework	Advantages of using the iOS app
Real-Time Notification Receiver	Real-Time Response
Receives Apple push notifications	Immediate response to threats.
Alerts IoT security expert researcher	Reduced response times.
Provides access to alarm information	Simplified validation process.

feedback and maintaining an indelible record of all security events within the IoT ecosystem. It ensures that the history of security incidents remains transparent and resistant to manipulation. Details are listed in Table 4.

4.1.5. Human expertise

Within the framework, Human Expertise represents the IoT security experts who play a crucial role in validating security alarms. These experts possess in-depth domain knowledge and expertise in identifying and addressing security threats. When alerted by the iOS Control Center, they diligently investigate the potential threat to determine its legitimacy. Their expertise is pivotal in distinguishing between genuine security threats and false positives.

4.1.6. Feedback loop

The Feedback Loop serves as a bridge between human expertise and blockchain technology. It enables IoT security experts to record their validated security incidents on the blockchain. This step is essential for preserving a comprehensive record of security events and expert assessments. The feedback loop ensures that the knowledge and insights of human experts are seamlessly integrated into the system.

4.1.7. Machine learning enhancement

The ML Enhancement component capitalizes on the wealth of feedback accumulated within the blockchain. ML algorithms systematically analyze this feedback to improve and adapt. The iterative process enables the algorithms to become increasingly accurate in identifying

Table 4
Role of blockchain in IoT security framework.

Blockchain benefits in IoT security	Description
Data Integrity and Immutability	Blockchain ensures the immutability and integrity of data records, preventing unauthorized alterations and deletions. This feature ensures the trustworthiness of security incident records.
Transparency and Trust	Blockchain's decentralized and transparent ledger fosters trust among IoT stakeholders. All participants can access and verify security incident details, ensuring transparency and accountability.
Security Incident History	Blockchain maintains a comprehensive historical record of security incidents and expert validations. This data is essential for trend analysis and threat detection.
Decentralization and Resilience	Decentralized blockchain networks enhance system resilience by eliminating single points of failure. Even if some nodes are compromised, the blockchain's integrity remains intact.
Smart Contracts for Automation	Blockchain platforms support smart contracts, enabling automated responses to security incidents. This feature enhances response times and system efficiency.

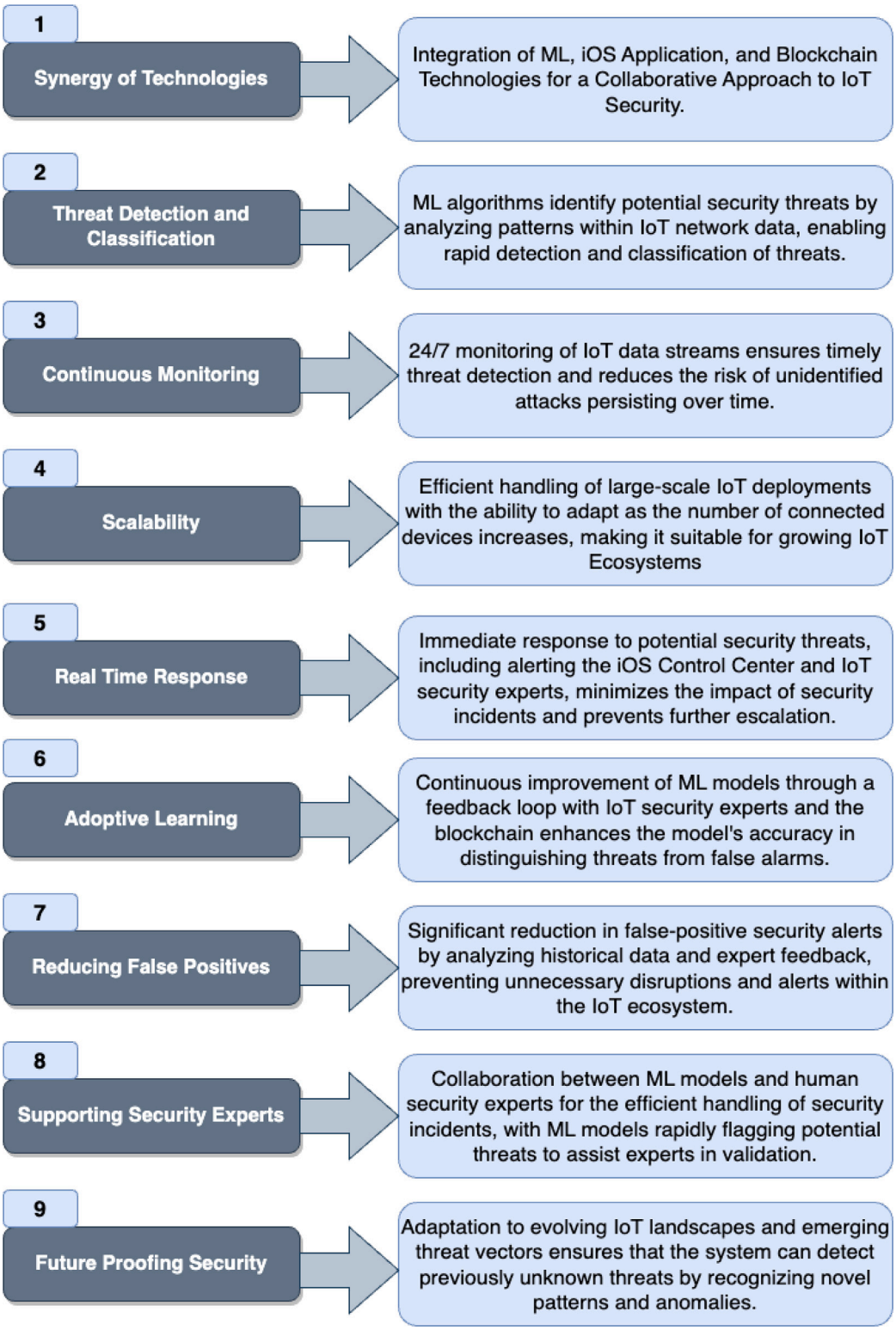


Fig. 3. Features of proposed framework.

true threats while minimizing false alarms. This enhancement is pivotal in refining the ML Layer's performance.

4.2. Information routing within the framework

The Next-Generation IoT Security Framework thrives on a well-orchestrated flow of information, fostering seamless communication and collaboration among its integral components. This framework is designed to enhance IoT security by unifying ML, an iOS Control Center, blockchain technology, and human expertise. Here, we elucidate the intricate information flow within this dynamic ecosystem.

The flow of data begins within the IoT Security Ecosystem, where an array of devices generates copious amounts of data and network traffic. This data stream forms the lifeblood of the framework, serving as the primary source of insights into potential threats. Continuously collected, this data is swiftly conveyed into the ML Layer. Here, a

Table 5
Flow of data within the IoT security framework.

Source	Destination
IoT Security Ecosystem (Data and Network Traffic)	Machine Learning Layer
Machine Learning Layer (Alert)	iOS Control Center
iOS Control Center (Push Notification)	IoT Security Expert Researcher
IoT Security Expert Researcher (Validation)	Feedback Loop (Blockchain)
Feedback Loop (Blockchain)	Machine Learning Layer
Machine Learning Layer (Model Improvement)	Machine Learning Layer

diverse set of algorithms takes center stage, analyzing incoming data in real-time. Their task is twofold: to identify potential security threats lurking amidst the data and to pinpoint deviations from established data patterns.

As soon as a potential threat is spotted, the ML Layer triggers an alert, marking the commencement of a collaborative chain reaction. This alert promptly flows to the iOS Control Center, serving as the communication hub of the framework. Here, the iOS Control Center dons multiple hats, acting as a messenger between the digital and human realms. It swiftly formulates an Apple push notification, packing it with relevant information about the potential threat, such as its nature and source. This notification is the baton passed to the designated IoT security expert researcher. Fig. 4 depicts the data flow process in detail where as Fig. 3 illustrates the characteristics of the IoT Security Framework.

Algorithm 1 Flow of Data Handling Algorithm

```
procedure HandleDataFlow
    Receive data from IoT Security Ecosystem
    Analyze data in Machine Learning Layer
    if Potential threat detected then
        Trigger alert
        Formulate push notification in iOS Control Center
        Send push notification to IoT security expert researcher
        IoT security expert researcher validates threat
        Record findings in Feedback Loop (Blockchain)
        Improve machine learning models
    end if
end procedure
```

With the push notification in hand, the IoT security expert researcher embarks on an investigative journey. The iOS Control Center plays a vital role, providing access to essential data and context surrounding the potential threat. Leveraging their domain knowledge and expertise, the human element assesses the legitimacy of the security alarm. This step is critical, as it determines whether the threat is genuine or a false positive, ensuring that the response is both timely and accurate.

Upon validation, the IoT security expert researcher records their findings within the system, marking an essential transition point to the Feedback Loop. This loop serves as a bridge, channeling human expertise and insights toward blockchain technology. It acts as the guardian of validated security incidents, ensuring that they are securely inscribed on the blockchain. The blockchain, in turn, plays a pivotal role, serving as an indelible ledger of security incidents, thus enhancing transparency and accountability.

The blockchain's repository of validated security incidents and expert assessments is not merely a historical archive. ML algorithms diligently examine this accumulated feedback, using it as nourishment for their growth. They continuously analyze, adapt, and evolve to improve their threat detection capabilities. This iterative learning process ensures that ML models become more adept at recognizing true threats from false alarms.

As these ML models are refined and honed through the feedback loop, their improvements cascade back to the ML Layer. Here, they are integrated into the algorithms, enhancing their accuracy and sharpening their capacity to distinguish genuine threats. This evolution of ML models contributes to proactive threat identification, strengthening the framework's security posture.

In essence, this continuous flow of information and the iterative improvement of ML models contribute to the overarching security evolution within the framework. The data flow is explained in detail in Table 5. Over time, the system becomes increasingly proficient at fortifying IoT security, with false alarms becoming rare occurrences. This evolving synergy between ML, human expertise, blockchain technology, and an iOS Control Center exemplifies the framework's potential to secure the ever-expanding IoT Security Ecosystem.

The Algorithm 2 offers a methodical strategy for enhancing security in IoT networks. It works in real-time and uses a variety of basic inputs, including ML classifiers for anomaly detection, input from an IoT security specialist, blockchain technology, and an iOS control centre. The ML classifiers immediately send a notification to the iOS control centre whenever they find an anomaly. The IoT security specialist then looks into the alarm. The expert's observations are added to the blockchain ledger, providing a transparent and immutable account of the security incident, if the anomaly is confirmed to be a real security challenge. The blockchain system incorporates this feedback over time in order to enhance ongoing learning processes.

5. Methodology

In the pursuit of bolstering IoT security, our research focuses on the collaborative integration of Blockchain and ML techniques. We aim to enhance threat detection and response mechanisms in IoT environments using advanced methodologies. Our study utilizes the IoT-23 Dataset to validate and evaluate the effectiveness of our proposed approach. Within this section, we expound upon the comprehensive approach undertaken to bolster the security of Internet of Things (IoT) systems. The methodology within this research embodies a cohesive blueprint, harmoniously integrating data preparation, ML algorithms, blockchain technology, and the creation of an iOS application. Through the harmonized interplay of these elements, we aim to cultivate a resilient and interconnected ecosystem that fortifies the very foundation of security within IoT landscapes.

5.1. Dataset

To train and evaluate our ML models, we utilized the IoT23 dataset, a widely recognized benchmark dataset for IoT security research. The IoT23 dataset comprises a diverse range of IoT-related data, including network traffic, device information, and various attack scenarios.

5.1.1. Data processing and pre-processing

In this section, we describe the crucial steps we undertook to ensure the quality, consistency, and relevance of our dataset. The process of data preprocessing and cleaning is essential for accurate and reliable results in any data-driven research project. To begin, we performed label cleaning, a vital step to accurately unify and categorize the numerous entries in our dataset. Many variations of labels existed, which required standardization for meaningful analysis. For instance, we transformed labels such as '- Malicious PartOfAHorizontalPortScan' and '(empty) Malicious PartOfAHorizontalPortScan' into the standardized label 'PartOfAHorizontalPortScan.' Additionally, we unified diverse representations of the word "Benign", such as "- Benign -", "(empty) Benign -", and "- benign -", to ensure consistency across the dataset. This meticulous label cleaning process resulted in a logically organized classification. Table 6 provides the label distribution after this cleaning process.

Algorithm 2 Advanced IoT Security Enhancement Framework

Input: Anomaly scores S from machine learning classifiers, IoT security expert E, blockchain technology B, iOS control center C.
Output: Enhanced IoT security, refined machine learning models M.
procedure OperateFramework
  while System operates in real-time **do**
    Calculate anomaly score s_i for incoming data stream D_i.
    if s_i exceeds threshold T **then**
      A = AnomalyDetection(D_i)
      if A is True **then**
        DispatchAlert(C)
        E investigates the alarm and retrieves data context DC from D_i.
        Compute severity score SS based on s_i and DC.
        if SS exceeds critical threshold CT **then**
          Expert E logs insights onto the blockchain B.
          B becomes a transparent, immutable record of security incidents.
        else
          Feedback is recorded for model refinement.
        end if
      end if
    end if
    Gather all feedback F from blockchain B.
    UpdateModel(M, F)
    Recompute threshold T based on new model M.
  end while
  Blockchain system B integrates expert feedback into ongoing learning processes.
  Machine learning models M adapt using both historical and real-time data.
  Framework optimizes T and minimizes false alarms, enhancing IoT security.
end procedure
function AnomalyDetection(D_i)
  Compute features f from D_i.
  Return True if $s_i(f) > T$, else return False.
end function
function DispatchAlert(C)
  Use push notification system of C to alert E.
end function
function UpdateModel(M, F)
  Incorporate F to retrain M.
end function

Table 6
Standardized label distribution after cleaning.

Label	Count
PartOfAHorizontalPortScan	685,184
Benign	490,967
C&C	14,958
DDoS	14,395
Attack	5962
C&C-Torii	30
C&C-File Download	14
File Download	3

Table 7
Final dataset classes after applying pre-processing steps.

Label	Count
Benign	490,967
PartOfAHorizontalPortScan	466,941
C&C	10,133
DDoS	9791
Attack	4070
C&C-Torii	22
C&C-File Download	7
File Download	3

Further preprocessing steps were undertaken to enhance the dataset's quality and relevance. Redundant columns were removed to improve the feature set. We ensured dataset integrity by detecting and eliminating duplicate records. The columns in the dataset include timestamp, source and destination IP addresses, ports, protocols, services, connection length, payload sizes, connection state, labels, and more.

Balancing the dataset was another crucial step to enhance the model's ability to generalize across different classes. This was achieved through the subsampling approach, resulting in a dataset comprising a total of 981,934 instances. Table 7 displays the label distribution in the final dataset after applying these preprocessing steps.

These meticulously preprocessed and standardized datasets lay the foundation for our subsequent analyses and model development. The well-balanced and refined dataset, as shown in Table 7, provides a fair representation of various attack types, ensuring the reliability and precision of our research findings.

Fig. 5 illustrates the resulting label distribution, corresponding to the data presented in Tables 6 and 7.

These standardized data preprocessing procedures serve as the foundation for our subsequent analyses and model development, guaranteeing the reliability and precision of our research findings. The next crucial stage is assigning numerical values to the output labels, which is essential for facilitating the training of ML models. This conversion simplifies the representation of various categories in an organized manner, making it easily understandable for ML algorithms to make predictions. As a result, the dataset labels are transformed as shown in Table 8.

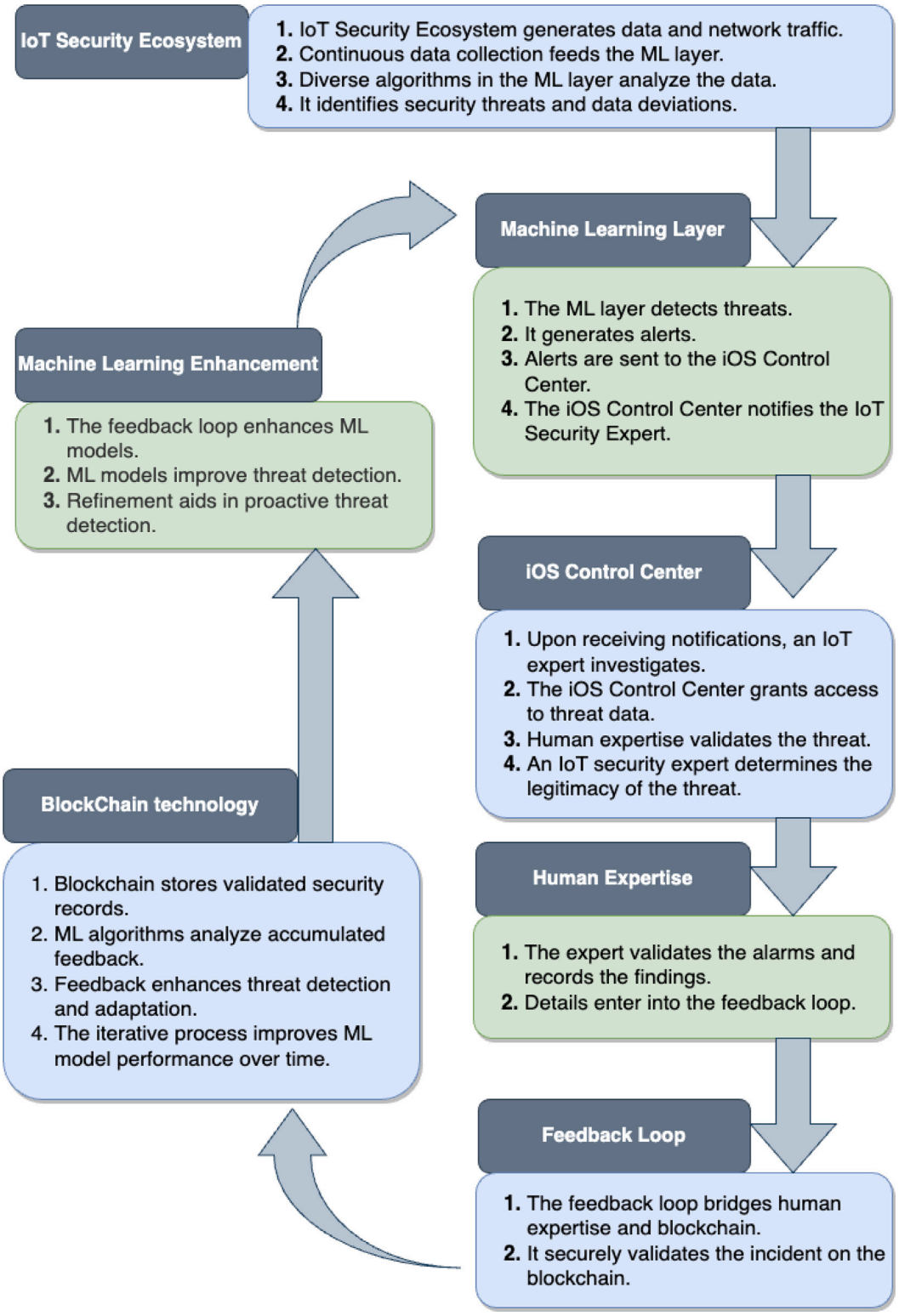


Fig. 4. Flow of data within IoT security frame work.

The dataset is now prepared for additional feature engineering, model creation, and analysis, thanks to the numerical encoding of the labels. This step is a crucial part of the overall process aimed at enhancing IoT security.

We used Incremental Principal Component Analysis (IPCA) in the dimensionality reduction stage of our methodology to convert the initial high-dimensional dataset into a lower-dimensional space while maintaining its key variability. Understanding how much each primary component contributes to the overall variance is possible thanks to the estimated explained variance ratios for each principal component. More specifically, the first principal component accounts for about 96.4% of the total variation, while the second component only makes up about 3.6%. The contributions of subsequent components are decreasing, and their explained variance ratios are getting less and smaller. With

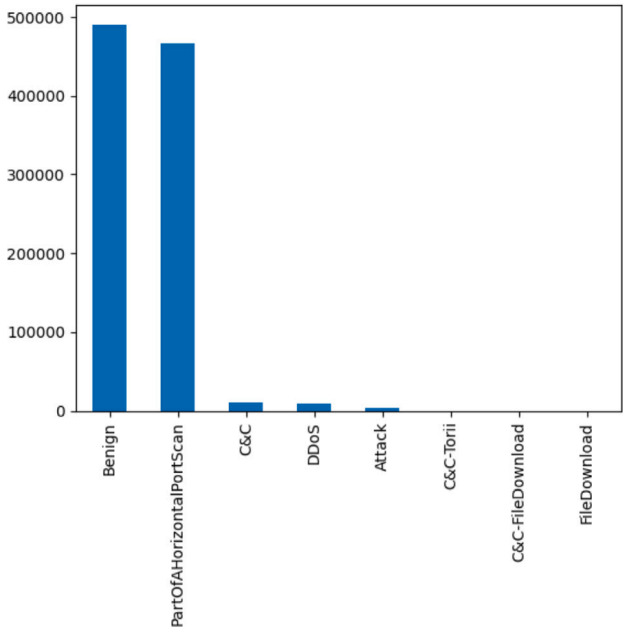


Fig. 5. Distribution of classes after data processing.

Table 8
Numerical encoding of dataset labels.

Original label	Numerical value
Benign	0
PartOfAHorizontalPortScan	1
Attack	2
DDoS	3
C&C	4
C&C-Torii	5
FileDownload	6
C&C-FileDownload	7

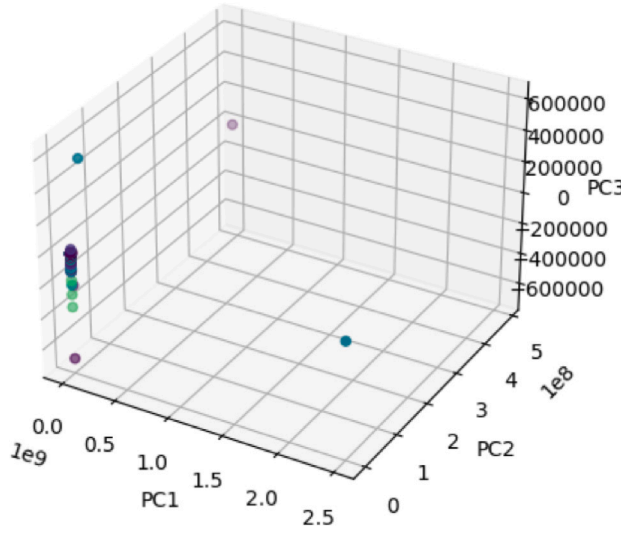


Fig. 6. 3D plot of incremental principal component analysis.

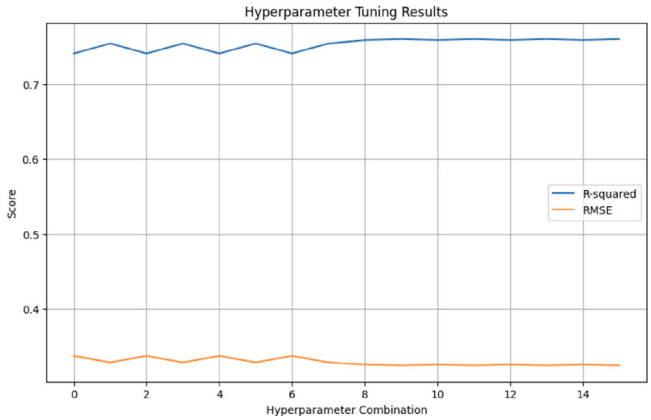


Fig. 7. HyperParameter tuning results.

the help of this analysis, we can decide how many main components to keep, making sure that the variance of the dataset is accurately represented in the reduced-dimensional space. The 3D scatter plot displayed in Fig. 8 presents a visual representation of data points in a three-dimensional space defined by the first three principal components (PC1, PC2, and PC3) obtained through PCA (Principal Component Analysis). This plot offers insights into the distribution and relationships among data points within this reduced feature space. As depicted in Fig. 6, the data points appear to form distinct clusters, suggesting that the reduced dimensions effectively capture patterns or groupings within the data. Outlying data points that deviate from these clusters are also visible, as potential outliers or unique data instances.

Post the fitting of the model to our dataset, we diligently calculated and documented the explained variance ratio for each principal component using the following equation:

$$\text{EVR} = \frac{\text{Variance of the Principal Component}}{\text{Total Variance in the Data}} \quad (1)$$

This ratio provided profound insights into the extent to which the variance in the original data was encapsulated by our reduced feature set.

In the methodology of our research, we employed hyperparameter tuning as a crucial step in optimizing the performance of our predictive model. Specifically, we applied hyperparameter tuning to fine-tune the hyperparameters of our XGBoostRegressor model. The hyperparameters under consideration included the number of estimators (`n_estimators`), maximum depth of trees (`max_depth`), minimum number of samples required to split an internal node (`min_samples_split`), and the maximum number of features to consider for each split (`max_features`).

To systematically explore the hyperparameter space, we utilized a grid search approach with cross-validation. The grid search involved an exhaustive search over predefined values for each hyperparameter, evaluating the model's performance using multiple evaluation metrics, including R-squared (`r2`) and negative root mean squared error (`neg_root_mean_squared_error`). The goal was to identify the optimal combination of hyperparameters that maximized the R-squared value, indicating the best fit of our model to the data. This meticulous process ensured that our model was fine-tuned to deliver the highest predictive accuracy. Figs. 7–9 depicts the Hyperparameter tuning results, best hyper parameters and models comparison before and after hyper tuning.

5.2. Machine learning algorithms

To address the unique challenges of IoT security, we selected a set of ML algorithms suitable for threat detection and classification. These algorithms include Random Forest, Decision Tree classifier, LSTM (Long Short-Term Memory), and CNN (Convolutional Neural Network). Each algorithm was implemented and configured based on its specific requirements and parameters. We divided the IoT23 dataset into training and testing sets to assess the performance of the ML models. The training set was used to train the ML algorithms using a supervised

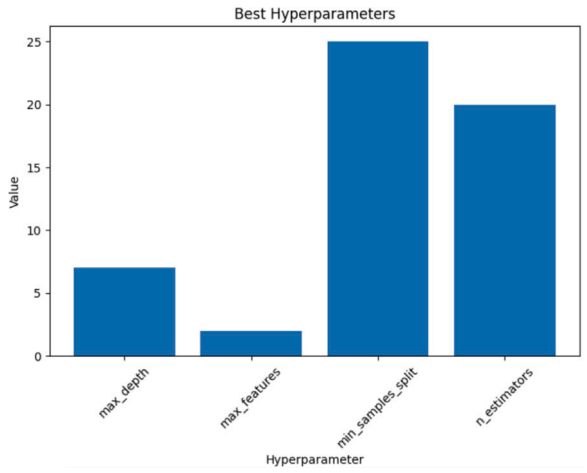


Fig. 8. Best hyper parameter.

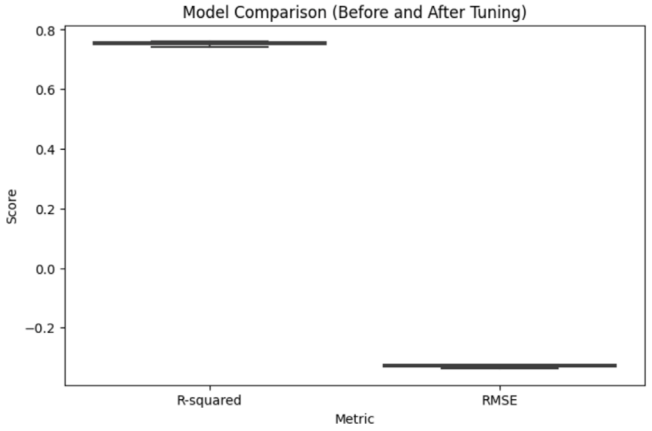


Fig. 9. Models comparison before and after tuning.

learning approach, where the features related to IoT security were used as input, and the corresponding threat labels were used as output. We employed cross-validation techniques to validate and optimize the models' hyperparameters, ensuring robustness and generalizability.

5.2.1. Performance metrics in collaborative threat intelligence

In our research on collaborative threat intelligence for IoT security, we employ a set of performance metrics to gauge the effectiveness of our approach. These metrics provide valuable insights into our system's ability to detect and respond to threats accurately. We use the IoT-23 Dataset to evaluate our collaborative threat intelligence system's performance.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{2}$$

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

$$F1\ \text{Score} = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{5}$$

Here, TP represents the number of correctly identified threats, TN stands for correctly identified non-threats, FP accounts for non-threats incorrectly classified as threats, and FN represents threats misclassified as non-threats.

Algorithm 3 Ensemble learning for Collaborative Threat Intelligence for IoT Security

Require: IoT-23 Dataset D, Split Ratio SR, Thresholds $threshold_malicious$, $threshold_benign$

procedure COLLABORATIVE_THREAT_INTELLIGENCE(D, SR, $threshold_malicious$, $threshold_benign$)
  Load and Pre-process IoT-23 Dataset:
    Perform essential data pre-processing steps on D, including cleaning and normalization.
  Split the Dataset:
    Randomly shuffle D.
    Calculate $split_index = \lfloor SR \times \text{size}(D) \rfloor$.
    $D_{\text{train}} = D[0 : split_index]$
    $D_{\text{test}} = D[split_index + 1 : \text{end}]$
  Train Ensemble Classifier:
    Initialize a Voting Classifier with multiple Machine Learning models.
    Train the Voting Classifier on D_{train}.
  Train Meta Ensemble Classifier:
    Initialize a Meta Ensemble model, e.g., GradientBoosting-Classifier.
    Train the Meta Ensemble model on predictions from base classifiers using D_{train}.
  Evaluate Model Performance:
    Initialize evaluation metrics variables.
    for each testing sample (x_j, y_j) in D_{test} **do**
      Extract features from x_j.
      $y_{\text{hat_ensemble}} = \text{VotingClassifier.predict}(x_j)$
      $y_{\text{hat_meta}} = \text{MetaClassifier.predict}(x_j)$
      Update evaluation metrics based on predicted and actual labels.
  Malicious/Benign Detection:
    for each testing sample (x_k, y_k) in D_{test} **do**
      Calculate $p_{\text{malicious_ensemble}} = \text{VotingClassifier.predict_proba}(x_k)$
      Calculate $p_{\text{malicious_meta}} = \text{MetaClassifier.predict_proba}(x_k)$
      if $p_{\text{malicious_ensemble}} > threshold_malicious$ **or** $p_{\text{malicious_meta}} > threshold_malicious$ **then**
        Generate alerts/notifications for potential IoT threats.
        Identify the source of threats using network analysis techniques.
        Implement appropriate security measures to mitigate attacks.
      if $p_{\text{malicious_ensemble}} < threshold_benign$ **and** $p_{\text{malicious_meta}} < threshold_benign$ **then**
        Continue with normal system operations.
  Output: Evaluation metrics, alerts/notifications, and actions taken for IoT security enhancement.
end procedure

These metrics enable us to comprehensively assess the performance of our collaborative threat intelligence system, ensuring accurate and reliable threat detection while minimizing false positives and false negatives.

5.2.2. Decision Tree classifiers

Our research harnesses the power of Decision Tree classifiers within the context of collaborative threat intelligence for IoT security. These classifiers are pivotal in analyzing the IoT-23 Dataset to detect and respond to threats effectively.

The Classification and Regression Tree (CART) algorithm, which constructs decision trees through recursive partitioning, is at the core of

our methodology. CART excels in segmenting data space into nodes for predictive modeling and evaluates node impurity using the Gini index. The Gini index for a node N is computed as:

$$\text{Gini}(N) = 1 - \sum_{c=1}^{C} (p_c)^2 \tag{6}$$

Where C represents the number of classes, and p_c signifies the proportion of class c instances within node N.

Our approach focuses on identifying optimal features F and thresholds T to minimize the Gini impurity Gini(D) while recursively descending through the decision tree. This construction process yields a comprehensive and predictive model for IoT threat detection, effectively capturing subtle patterns in the IoT-23 Dataset.

5.2.3. Naïve Bayes

Our research employs Naïve Bayes classifiers to enhance IoT threat detection within the collaborative threat intelligence framework. Naïve Bayes classifiers are known for their simplicity and effectiveness in handling diverse datasets, making them well-suited for analyzing the IoT-23 Dataset.

The algorithm leverages Bayes' theorem to calculate probabilities and classify data effectively. It provides probabilistic predictions and is particularly useful in distinguishing between regular and threat instances.

$$P(H \mid x) = \frac{P(x \mid H)P(H)}{P(x)} \tag{7}$$

Where $P(H \mid x)$ is the probability of a threat given the input data, $P(x \mid H)$ is the probability of the input data given a threat, $P(H)$ is the prior probability of a threat, and $P(x)$ is the prior probability of the input data.

Naïve Bayes classifiers enable our collaborative threat intelligence system to make informed decisions based on probabilistic analysis, contributing to accurate threat detection and response.

5.2.4. Random forest classifier

In our collaborative threat intelligence research, we incorporate the Random Forest Classifier as an ensemble learning technique to improve threat detection using the IoT-23 Dataset. This ensemble method combines predictions from multiple decision trees, enhancing the accuracy and robustness of our threat detection system.

By training on bootstrap samples and using a voting mechanism to aggregate predictions, the Random Forest Classifier effectively minimizes overfitting and improves generalization. This makes it a valuable tool in identifying malicious activity within IoT networks.

Our research demonstrates the power of the Random Forest Classifier in enhancing IoT threat detection. By leveraging the strengths of multiple decision trees, our system achieves higher accuracy and reliability in identifying and responding to threats effectively.

5.2.5. Voting classifier in ensemble learning

Our collaborative threat intelligence research introduces an ensemble model that integrates Blockchain and ML techniques within the Voting Classifier framework. This ensemble approach combines the advantages of various ML models, contributing to improved IoT threat detection and response.

$$\begin{aligned}&\text{classifier_model} = \text{VotingClassifier}\\&(\text{estimators} = [(\text{`}nb\text{'}, \text{GaussianNB}()),\\&((\text{`}dt\text{'}, \text{DecisionTreeClassifier}()),\\&(\text{`}rf\text{'}, \text{RandomForestClassifier}())],\\&\qquad\qquad \text{voting} = \text{`}hard\text{'})\end{aligned} \tag{8}$$

The Voting Classifier employs a majority voting strategy to aggregate predictions, enhancing our system's accuracy and reliability. By unifying Naïve Bayes, CART, and Random Forest algorithms within this ensemble framework, our research ensures effective IoT threat detection, contributing to enhanced IoT security. Algorithm 3 outlines the procedures involved in training and testing our Collaborative Threat Intelligence Model, providing a comprehensive overview of our approach.

5.2.6. Long Short-Term Memory (LSTM)

In our research, we harnessed the capabilities of LSTM networks to enhance the security framework for IoT. LSTM networks, designed to capture sequential dependencies in data, are particularly suited for processing time-series IoT data efficiently. Our model consists of multiple LSTM layers, each with its own activation functions and dropout layers to mitigate overfitting. Mathematically, the LSTM operation involves a series of computations, including the application of hyperbolic tangent (tanh) activation functions and sigmoid functions to gate input information and control the flow of data through time steps. By stacking these LSTM layers, we create a deep learning architecture capable of capturing and learning complex patterns in the IoT data. The LSTM model is trained using categorical cross-entropy loss and optimized with the Adam optimizer over 25 epochs. This application of LSTM networks showcases their potential in enhancing the precision and performance of our IoT security framework, ultimately contributing to the broader objectives of our research.LSTM layers are defined as follows:

$$h_t = \tanh(W_{ih}x_t + b_{ih} + W_{hh}h_{t-1} + b_{hh}) \tag{9}$$

$$\text{Output gate: } o_t = \text{sigmoid}(W_{io}x_t + b_{io} + W_{ho}h_{t-1} + b_{ho}) \tag{10}$$

$$\text{Cell gate: } \tilde{C}_t = \tanh(W_{ic}x_t + b_{ic} + W_{hc}h_{t-1} + b_{hc}) \tag{11}$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \tag{12}$$

Where: - h_t represents the hidden state at time t. - x_t is the input at time t. - o_t is the output gate. - $\tilde{C}_t$ is the cell gate. - C_t is the cell state. - f_t is the forget gate. - i_t is the input gate. - W and b are weights and biases.

5.2.7. Convolutional Neural Network for IoT security

In our quest to fortify our IoT security framework, we harnessed the formidable capabilities of a CNN, meticulously designed for processing time-series IoT data. This specialized CNN excels in unveiling intricate patterns inherent in our dataset. The model encompasses multiple layers, commencing with three convolutional layers, each succeeded by strategic max-pooling operations for feature down-sampling. Additionally, dense layers, fortified with dropout mechanisms to mitigate overfitting risks, are seamlessly integrated into the architecture. Our comprehensive CNN leverages fundamental mathematical operations, specifically convolution (∗) Eq. (13), pooling (Max-Pooling) Eq. (14), and activation functions (ReLU) Eq. (15), working in synergy to extract and distill salient features from our IoT data. The model undergoes meticulous training, optimizing its performance through categorical cross-entropy loss and harnessing the Adam optimizer. Over the course of 15 epochs, our CNN adeptly categorizes IoT data into distinct threat categories, underscoring the transformative potential of deep learning within the realm of IoT security, thus solidifying its profound implications for our collaborative research.

$$(f * g)(n) = \sum_{m=-\infty}^{\infty} f(m)g(n-m) \tag{13}$$

$$\text{Max-Pooling}(x) = \max(x) \tag{14}$$

$$\text{ReLU}(x) = \max(0, x) \tag{15}$$

5.3. Integration of iOS application

At the heart of our IoT security framework resides the iOS application, which functions as a dynamic control center. The applica-

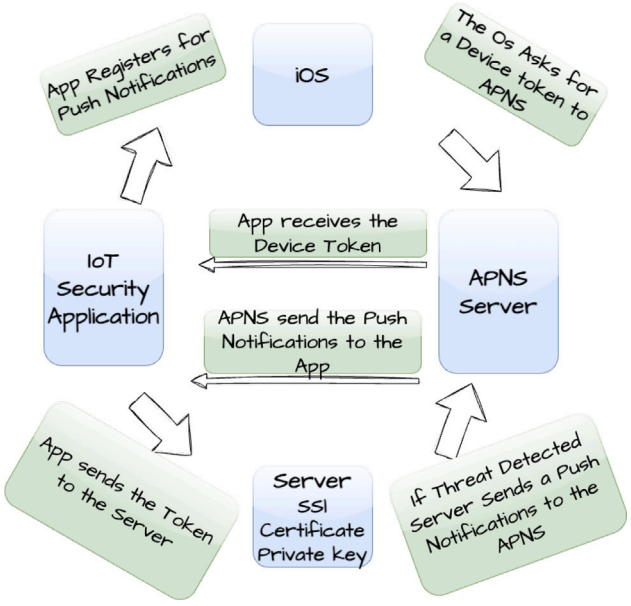


Fig. 10. How do iOS Push Notifications work?.

tion's interface facilitates intuitive navigation for users to oversee and manage IoT security parameters. Additionally, real-time threat alerts are seamlessly generated, drawing insights from our machine-learning models. Apple Push notification system empowers security experts to respond promptly to potential breaches. Moreover, the iOS application seamlessly interacts with blockchain technology, facilitating secure submission and retrieval of feedback through smart contracts. For secure transmission of threat data, we established encrypted communication channels between the iOS application and the blockchain network. We utilized smart contracts to securely store and share threat information on the blockchain, ensuring the immutability and integrity of the data. The iOS application was integrated with the ML models, enabling real-time threat detection and classification. While Developing iOS Application we have followed standards for the security of iOS Application, As shown in Table 9 and Fig. 11 (Apple Inc., 2023).

One of the key features of our iOS application is its real-time threat alert system, powered by the Apple Push Notification service. This robust notification system enables security experts to receive immediate alerts in response to potential breaches, ensuring rapid and effective responses. By harnessing the power of push notifications, our application enhances the timeliness and efficiency of threat detection and response, further fortifying the security of IoT ecosystems. To implement Apple Push Notifications within our iOS application, we followed a structured approach. First, we integrated the Apple Push Notification service (APNs) into our application, utilizing Apple's official libraries and APIs. Next, we established a secure connection between our application server and APNs using Transport Layer Security (TLS) to ensure data privacy and integrity during transmission. We configured the necessary certificates and provisioning profiles to enable secure communication with APNs. Subsequently, we implemented logic within our application to trigger push notifications in real-time upon the detection of potential security threats by our ML models. These notifications were designed to provide security experts with detailed information about the detected threat, enabling them to take swift and informed actions. By adhering to Apple's guidelines and best practices, we ensured a robust and reliable implementation of Apple Push Notifications, a crucial component of our IoT security framework. Algorithm 4 and Fig. 10 portray the essential procedures for the deployment of push notifications through APNs within the context of our IoT security framework.

Algorithm 4 Push Notification via APNs

Initialization:
Set up Apple Developer account.
Generate APNs Key.
Configure Xcode project.
Server-Side:
Implement secure channel.
Set up server/cloud service.
User Registration:
Register device token.
Store token with user info.
Real-Time Detection:
Implement ML models.
Trigger server notification.
Notification Request:
Create JSON payload.
Send Notification:
Use APNs API.
Include JSON payload.
Display:
iOS displays notification.
Ensure clarity.
User Interaction:
Enable user response.
Monitoring:
Monitor delivery status.
Handle failures.
Security and Privacy:
Encrypt transmission.
Ensure compliance.
Testing and Optimization:
Test thoroughly.
Optimize for scalability.
Documentation:
Document process.
Maintain logs.

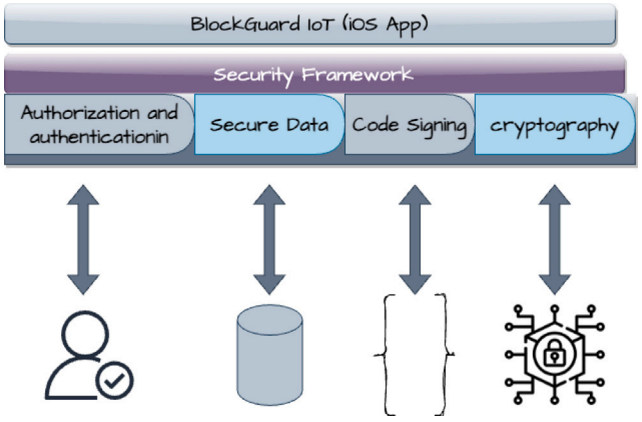


Fig. 11. iOS application security framework.

5.4. Blockchain integration for immutable feedback and ML model enhancement

In this section, we explore the nuanced details of our methodology, focusing on the seamless integration of blockchain technology into our research framework. The primary objective of this integration is to establish a resilient and immutable feedback system within our IoT security solution. Blockchain plays a pivotal role in the stability and evolution of our IoT security framework, serving as the cornerstone for

Table 9
Security measures and descriptions.

Security measure	Description
Secure Code Development	We rigorously adhered to secure coding practices and incorporated trusted libraries during the iOS application development process. This ensured that the codebase was fortified against common vulnerabilities and potential exploits.
Data Encryption	Sensitive data, both in transit and at rest, underwent robust encryption mechanisms. By employing industry-standard encryption protocols, we ensured that confidential information remained safeguarded even in the face of unauthorized access.
Authentication and Authorization	The iOS application implemented robust user authentication and authorization mechanisms. This ensured that only authorized users could access specific functionalities and data, enhancing overall app security.
Secure User Inputs	Stringent user input validation and sanitization measures were put in place. This thwarted potential injection attacks and malicious inputs, bolstering the application's resilience against such threats.
Secure Data Storage	The storage of sensitive data within the application was fortified through secure storage techniques. This ensured that critical information remained protected from unauthorized access, even in the event of device compromise.
Secure Network Communication	The application exclusively employed secure HTTPS communication protocols, further fortified by implementing certificate pinning. This thwarted potential eavesdropping and man-in-the-middle attacks.
App Transport Security (ATS)	The application's network communication was configured in adherence to App Transport Security guidelines. This ensured that network connections were secure, reducing the potential for data interception.
Secure APIs	We implemented secure authentication mechanisms for APIs, preventing unauthorized access and data breaches through API endpoints.

Table 10
Smart contracts for research framework.

Smart contract name	Description
Device Identity and Authentication	Manages the creation and verification of unique digital identities for IoT devices.
Data Encryption	Handles encryption and decryption of IoT data to ensure confidentiality and integrity.
Audit Logging	Records all changes to IoT data and device identities for traceability and security incident investigations.
Feedback Recording	Automates the recording of feedback from IoT security experts, securely stored on the blockchain.
Consensus Mechanism	Manages the consensus process for verifying feedback and transactions (platform-specific, e.g., Ethereum).
Resource Allocation	Allocates computational resources for training and deploying machine learning models fairly.
Tokenization	Facilitates token transactions and balances for token-based incentives or payments.
Machine Learning Model Management	Oversees deployment and updates of machine learning models on the blockchain.
User Access Control	Controls access permissions and rights to specific system components or data.
Alert Generation and Propagation	Generates security alerts based on machine learning model outputs and propagates them to stakeholders.

a transparent and robust feedback loop that enhances the accuracy of the ML model's alert generation. This section accentuates the pivotal functions of blockchain while delving into the comprehensive methodology that underpins its integration into our system. In-depth insights into Blockchain features and protocols are elaborated upon in Fig. 12, Tables 11 and 12.

5.4.1. Verification and feedback loop

Our methodology seamlessly integrates a sophisticated verification and feedback loop, harnessing the synergies between automated alert generation and expert human analysis. When the ML model identifies anomalies within the IoT data stream, it promptly initiates an alert. This alert serves as a trigger for a subsequent validation phase, in which IoT security experts, equipped with profound domain-specific insights, meticulously scrutinize and authenticate the generated alert. This meticulous verification procedure introduces a pivotal stratum of human intelligence into the system, serving to significantly diminish the likelihood of spurious alarms and concurrently elevating the accuracy and precision of our security alert mechanisms.

5.4.2. Blockchain-based feedback

Following a meticulous verification process by IoT security experts, an extensive and detailed record of the verification, incorporating supplementary feedback and context, is systematically and immutably archived within the blockchain infrastructure. The blockchain ledger, distinguished for its indisputable immutability, assumes the role of an impervious repository meticulously safeguarding these records. This blockchain-driven feedback mechanism operates as a repository of the combined proficiency of the security professionals, rendering it an invaluable repository of insights and contextual information. These reservoirs of knowledge are poised to serve as an enduring foundation for the forthcoming alert generations, enriching the ML model's

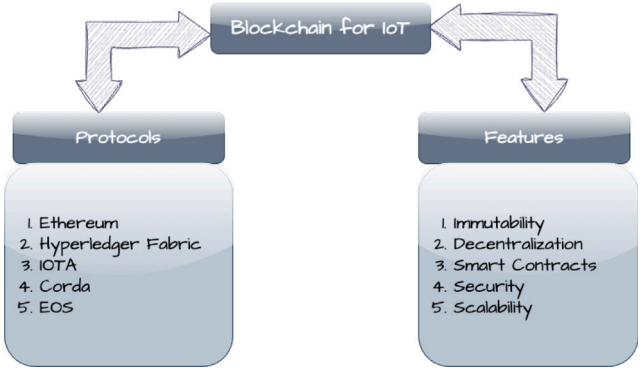


Fig. 12. Block chain for IoT, protocols, and features.

anomaly detection capabilities through the incorporation of collective security expertise. This intricate feedback system underscores the pivotal role played by blockchain technology in enhancing the resilience and effectiveness of our IoT security framework.

5.4.3. ML model enhancement

The blockchain's influence transcends the realm of storage; it plays an active role in elevating the proficiency of our ML model. Periodic scrutiny of verified alerts and feedback is conducted by a dedicated ML Enhancement component. This analytical process equips the ML model with the ability to iteratively adjust and enhance its anomaly detection algorithms over time. The net result is a discernible reduction in the occurrence of false alarms and a substantial enhancement in the precision and dependability of our security alerting system. This integration not only bolsters the reliability of our framework but also underscores the pivotal role of blockchain technology in fostering a continually evolving and adaptive IoT security environment.

5.4.4. Immutable records and prevention of false alarms

The immutability of blockchain records ensures that feedback and verification results remain unaltered, preserving their integrity and authenticity. By harnessing the collective intelligence stored on the blockchain, our framework continually enhances its ML models. This proactive approach prevents false alarms and provides more accurate security alerting, thereby significantly bolstering the security of interconnected IoT ecosystems.

5.4.5. Blockchain integration for enhanced IoT security

The implementation of blockchain technology within our research framework represents a critical strategic choice, one that significantly augments the robustness and efficacy of our IoT security solution. By selecting a private blockchain platform, we have gained invaluable control over the blockchain's operation, ensuring that it seamlessly aligns with the unique security prerequisites of our research. This strategic decision enables us to capitalize on specific blockchain features and protocols, each of which plays a pivotal role in enhancing the security, transparency, and adaptability of our IoT security ecosystem.

Central to our blockchain integration are smart contracts, multifunctional digital agreements that automate key processes within our framework. These intelligent contracts serve as the backbone of our feedback loop, orchestrating the entire feedback lifecycle, from recording to verification and enhancement. Smart contracts facilitate frictionless interactions between different components of our ecosystem, making the feedback process efficient, transparent, and tamper-proof.

Our methodology harnesses the power of a distributed ledger, the foundational technology behind blockchain's immutability. This distributed ledger serves as a secure repository, housing a comprehensive record of verified alerts, feedback, and the outcomes of the verification process. Importantly, it maintains an immutable ledger of the entire security alerting process, ensuring that records remain tamper-proof and transparent. This feature is pivotal in maintaining the integrity of the feedback loop.

Table 11
Blockchain features for IoT.

Feature	Description
Immutability	Ensures data integrity and tamper resistance.
Decentralization	Eliminates single points of failure.
Smart Contracts	Enables automated and self-executing agreements.
Security	Provides robust cryptographic security.
Scalability	Handles vast amounts of IoT data efficiently.

The utilization of cryptographic techniques is a cornerstone of our blockchain implementation. Robust cryptographic measures are meticulously employed to secure both the storage and transmission of sensitive data within the blockchain. This cryptographic layer safeguards feedback data and verification outcomes, ensuring their confidentiality, integrity, and authenticity throughout their lifecycle on the blockchain.

The strategic incorporation of blockchain technology within our research methodology not only fortifies the security and transparency of our IoT security framework but also actively contributes to the enhancement of our ML models. The blockchain-enabled feedback loop, verification process, and ML model enhancement collectively ensure a more accurate, reliable, and adaptive security alerting system. This integrated approach positions our research to effectively prevent false alarms, swiftly adapt to emerging threats, and elevate the security of interconnected IoT ecosystems.

In our research structure, the adoption of blockchain technology is anchored in the extensive use of diverse smart contracts, each crafted to serve a distinct role within the IoT security framework. As showcased in Table 10 and outlined in Algorithms 5, 6, 7, and 8, these smart contracts deliver a broad spectrum of capabilities, such as device identification and authentication, data ciphering, audit trail creation, and feedback documentation. In addition, smart contracts preside over fundamental procedures like consensus determination, resource distribution, and management of ML models. Acting as the foundational pillar of our IoT security strategy, these blockchain-fortified smart contracts assure clarity, traceability, and seamless execution of vital security tasks. Their significance is not confined to mere data documentation; they proactively bolster the efficiency and fortification of the overall system, a theme expanded upon in the following sections.

Algorithm 5 IoT Device Registration Smart Contract

```
procedure RegisterDevice(_deviceId (string), _deviceType (string))
    Input: _deviceId (string), _deviceType (string)
    Output: None

    if bytes(_deviceId).length > 0 then
        if bytes(_deviceType).length > 0 then
            Create a new device with the following attributes:
                - owner: msg.sender
                - deviceId: _deviceId
                - deviceType: _deviceType
                - timestamp: block.timestamp
            Add the new device to the owner's list of devices.
            Emit a DeviceRegistered event with the following parameters:
                - owner: msg.sender
                - deviceId: _deviceId
                - deviceType: _deviceType
                - timestamp: block.timestamp
        end if
    end if
end procedure
```

Algorithm 6 User Registration Smart Contract

```
procedure RegisterUser
    Input: None
    Output: None

    if registeredUsers[msg.sender] == false then
        Register the user by setting registeredUsers[msg.sender] to true
        Emit a UserRegistered event with the following parameter:
          - user: msg.sender
    end if
end procedure
```

Algorithm 7 User Authentication Smart Contract

```
procedure AuthenticateUser(_user (address))
    Input: _user (address)
    Output: None

    if UserRegistry(msg.sender).isUserRegistered(_user) == true then
        if authenticatedUsers[_user] == false then
            Authenticate the user by setting authenticatedUsers[_user] to true
            Emit a UserAuthenticated event with the following parameter:
              - user: _user
        end if
    end if
end procedure
```

Algorithm 8 Feedback Submission Smart Contract

```
procedure SubmitFeedback(_feedbackText (string))
    Input: _feedbackText (string)
    Output: None

    Create a new feedback with the following attributes:
      - securityExpert: msg.sender
      - feedbackText: _feedbackText
      - timestamp: block.timestamp
    Add the new feedback to the user's feedback list.
    Emit a UserFeedbackReceived event with the following parameters:
      - user: msg.sender
      - feedbackText: _feedbackText
      - timestamp: block.timestamp
end procedure
```

Table 12
Blockchain protocols for IoT.

Protocol	Description
Ethereum	Popular platform with smart contract capabilities.
Hyperledger Fabric	Permissioned blockchain for enterprise use.
IOTA	Designed specifically for IoT with feeless transactions.
Corda	Distributed ledger with a focus on financial services.
EOS	Known for scalability and high transaction throughput.

5.5. Systems and technologies utilized for development

Our study, "Collaborative Threat Intelligence: Enhancing IoT Security through Blockchain and Machine Learning Integration", presents a thorough framework made to improve IoT security. This architecture has three crucial elements: blockchain technology, ML, and an iOS application acting as a central command post. Different tools and technologies are used to fully utilize the capabilities of this framework. These include blockchain platforms for building secure and transparent ledgers, ML frameworks and libraries for establishing predictive models, and iOS development tools for producing the control center application. The effective handling of the growing internet of things security issues by our study is made possible by the seamless integration of these tools and technologies, which ensures the integrity, confidentiality, and availability of connected devices and their data. Table 13 outlines the resources needed for this research project.

Table 13
System and technology usage.

System/Technology	Purpose/Usage
Development Workstation	Primary environment for coding and development, Intel Core i7 processor, 16GB RAM, SSD storage
Xcode & MacBook Air	Creation, testing, and refinement of iOS application
iPhone Device	Real-world compatibility and functionality testing
Cloud-Based Infrastructure	Deployment of blockchain nodes and network interaction
Cloud Services	Resource scaling for blockchain network and machine learning
Git & GitHub	Version control and collaborative development
Machine Learning Platforms	Cloud platforms for training and fine-tuning ML models

In order to increase the security of IoT systems, this methodology emphasizes the integration of ML, blockchain, and iOS application control. Our strategy adds to a comprehensive IoT security solution through the inclusion of real-time threat detection, open feedback channels, and user-friendly controls. By integrating these components together, we develop a complete security solution that not only secures IoT systems but also provides users with useful information for proactive threat mitigation. The foundation of the next-generation IoT security paradigm is the integration of blockchain, ML, and iOS application control.

5.6. Computational and communication complexities in implementation

The integration of Blockchain, Machine Learning (ML), and the iOS application into a collaborative threat intelligence framework for enhancing IoT security introduces a myriad of computational and communication complexities. A comprehensive understanding and effective resolution of these challenges are crucial to ensuring the efficiency and reliability of the developed system.

5.6.1. Blockchain complexity

The incorporation of Blockchain technology introduces intricate computational and communication challenges, predominantly tied to the consensus mechanism, transaction verification, and distributed ledger management. The decentralized nature of Blockchain necessitates substantial computational resources to achieve consensus among network nodes. Furthermore, ensuring secure and efficient communication between nodes within the Blockchain network adds an additional layer of complexity.

5.6.2. iOS application (Swift) complexity

Developing the iOS application using Swift entails addressing challenges associated with user interface design, data communication with the backend (Blockchain and ML components), and ensuring compatibility with a diverse range of iOS devices. The complexities also extend to incorporating robust security features within the application, such as implementing secure communication protocols and seamless integration with the Apple Push Notification Service (APNS).

Table 14
Computational and communication complexities in implementation.

Category	Complexities
Blockchain	Consensus mechanism, transaction verification, decentralized ledger management, secure inter-node communication.
iOS Application (Swift)	User interface design, backend communication, device compatibility, security features, integration with APNS.
Machine Learning	Model training, real-time inference, continuous learning, feedback loop integration with Blockchain.

5.6.3. Machine learning (ML) complexity

The integration of ML models into the collaborative framework necessitates addressing computational complexities related to training, inference, and continuous learning. Training sophisticated ML models on large datasets demands substantial computational resources. Real-time processing of IoT data streams for inference introduces challenges in terms of latency and computational efficiency. Additionally, establishing a seamless feedback loop for continuous model improvement involves addressing communication complexities between the ML component and the Blockchain network.

The comprehensive overview of the computational and communication complexities is presented in Table 14, highlighting the multifaceted challenges in each category.

Addressing these complexities necessitates a holistic approach, considering the intricate interplay of Blockchain, Swift, and ML components to achieve a robust and efficient collaborative threat intelligence framework for enhancing IoT security.

6. Result and analysis

In accordance with the research study, this section elucidates the findings arising from our comprehensive exploration aimed at fortifying IoT security. The confluence of an iOS application functioning as a control center, ML techniques for discerning patterns, and the incorporation of blockchain technology with smart contracts forms the core of our investigation. Our evaluation is firmly grounded in the research objectives, encompassing meticulous scrutiny of diverse ML models across the expansive IoT23 dataset, housing an array of both benign and malicious activities. Within this analysis, we scrutinize the standalone efficacy of each model as well as undertake a comparative appraisal, thereby shedding light on their precision in effectively distinguishing between distinct IoT data classes. Furthermore, we navigate the realm of blockchain technology and smart contracts, expounding on how this cryptographic fusion augments data storage, security, and transparency within the IoT ecosystem. By systematically delineating the ramifications of our experimentation, we intend to proffer insightful revelations into the symbiotic potential of these integrated technologies in fortifying IoT security.

6.1. Machine learning model performance

In this section, we conducted a meticulous assessment of five powerful ML models: Random Forest, Decision Tree, Naive Bayes, and two deep learning classifiers, Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM). Additionally, we explored the performance of an ensemble classifier that combines the strengths of these individual models. Our primary objective was to address the critical challenge of threat detection within IoT ecosystems. Throughout our experimentation, we assigned numerical labels to distinct threat classes, ensuring that our models could effectively distinguish between benign network activities and various types of threats.

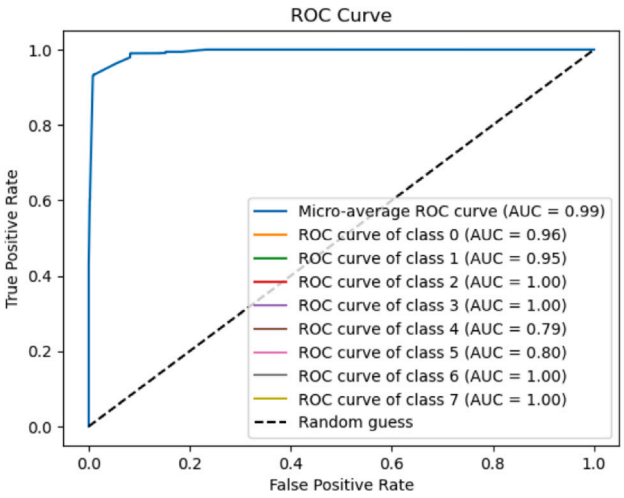


Fig. 13. RoC curve for random forest classifier.

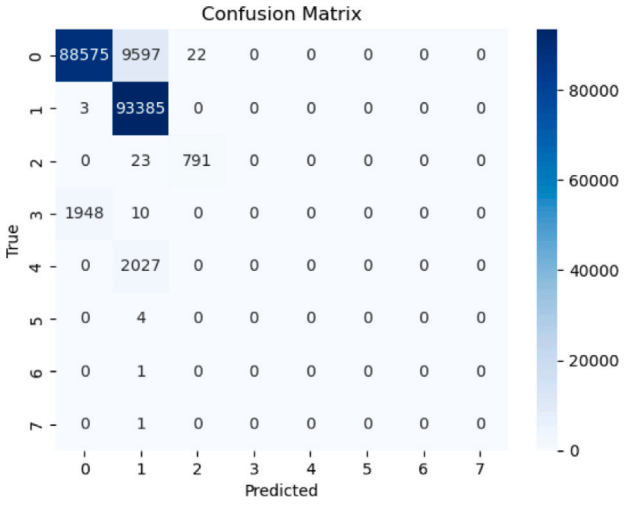


Fig. 14. Confusion matrix for random forest classifier.

6.1.1. Performance of the random forest classifier

The performance of the Random Forest classifier yielded highly promising results. Receiver Operating Characteristic (ROC) curves for each class showcased the model's exceptional discrimination capabilities. Impressively, classes representing severe threats, such as Distributed Denial of Service (DDoS) and Command and Control (C&C) activities, exhibited Area Under the Curve (AUC) values of 1.00, indicating impeccable classification. This underscores the Random Forest's proficiency in accurately identifying and distinguishing malicious network behaviors. Furthermore, our evaluation metrics, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R-squared, and Mean Absolute Error (MAE), pointed to robust model performance with minimal prediction errors. Notably, our Random Forest model demonstrated a strong correlation between predicted and actual outcomes, with a training R-squared value of 50.1% and a testing R-squared value of 49.9%. These findings enhance the credibility of our research. Figs. 13–18 depict the performance of the Random Forest Classifier on the IoT23 dataset for our collaborative research.

6.1.2. Decision Tree classifier

Shifting focus to the Decision Tree classifier, our findings provided valuable insights. The ROC curves illustrated the model's effectiveness in distinguishing different threat classes, with AUC values ranging from

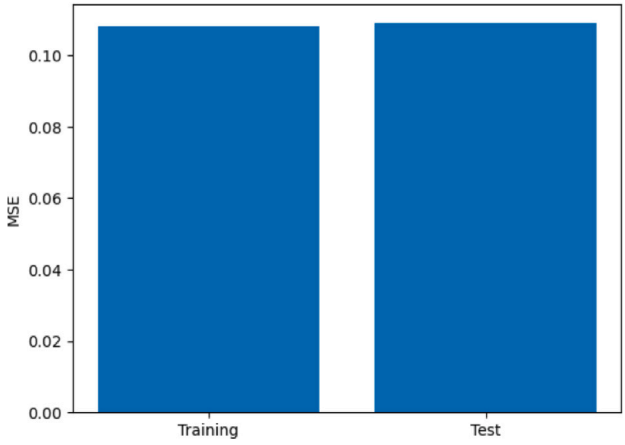


Fig. 15. MSE training and testing Random Forest.

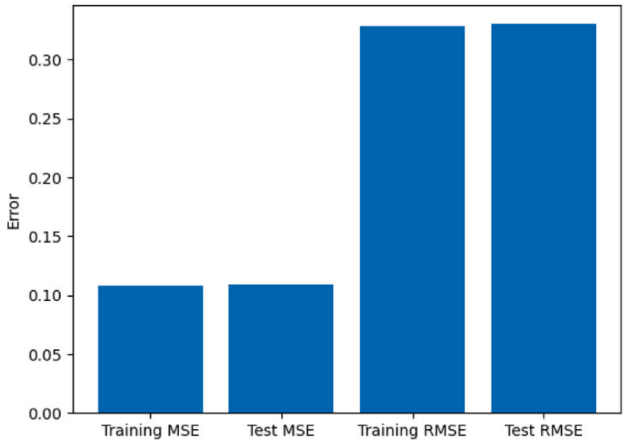


Fig. 16. Training and Test Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) for Random Forest.

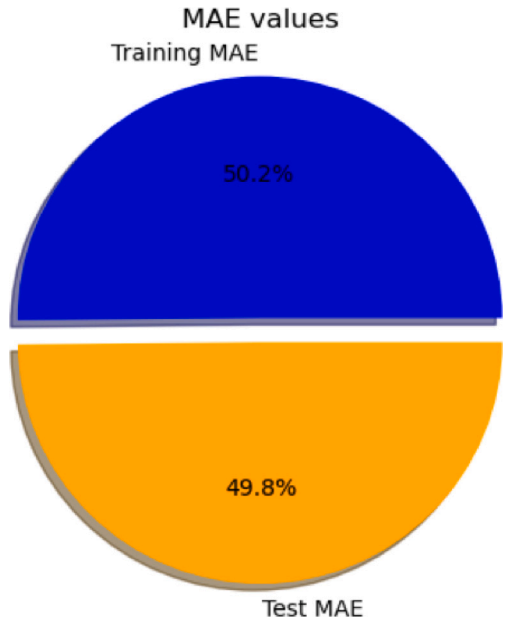


Fig. 17. MAE values of Random Forest for training and testing.

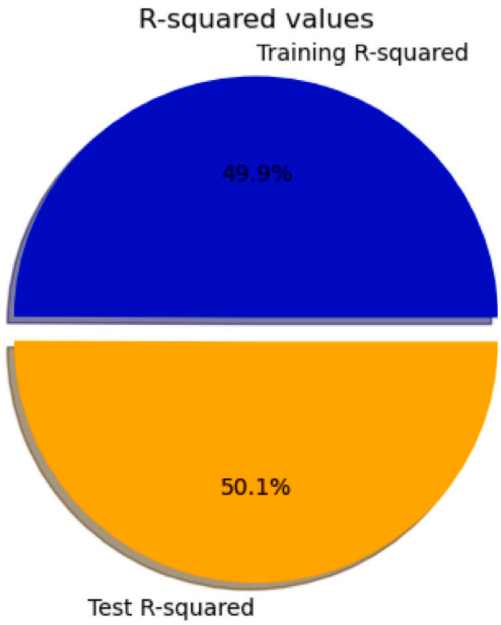


Fig. 18. R-Squared of Random Forest for training and testing.

1.00 to 0.95. Once again, severe threats like DDoS and C&C activities achieved perfect AUC scores, validating the model's capability to identify high-risk network activities. Regarding evaluation metrics, the model displayed competitive performance, as reflected in MSE, RMSE, R-squared, and MAE values. A training R-squared value of 50.3% and a testing R-squared value of 49.7% emphasized the model's aptitude for capturing underlying data patterns. MAE values of 49.3 for training and 50.7 for testing demonstrated its ability to make precise predictions while maintaining errors within acceptable bounds. Fig. 19 displays the Tree Visualization for the decision tree classifier, while Fig. 20 illustrates the ROC curve for the Decision Tree Classifier when applied to the IoT-23 Dataset.

In our research, the process of feature engineering played a pivotal role in enhancing the effectiveness of our IoT security framework. Feature engineering involves selecting and transforming relevant data attributes to create new features that better represent the underlying patterns in the data. One of the key outcomes of our feature engineering efforts was the identification of the most important features contributing to the accuracy and performance of our system. To quantify the significance of each feature, we employed a Decision Tree classifier, a powerful ML algorithm capable of assessing feature importance. The algorithm evaluated the potential of each feature to discriminate between different security states and threats within our IoT ecosystem. The result was a set of feature importance scores, each indicating the degree to which a specific attribute influenced our framework's decision-making process.

Mathematically, the feature importance scores (FIS) can be represented as follows:

$$FIS_i = N \sum_{j=1}^{N} |C_i - C_{ij}|$$

Where:

FIS_i represents the feature importance score of feature i.

C_i is the accuracy of the model using all features.

C_{ij} is the accuracy of the model after removing feature i.

N is the total number of features.

Our findings revealed a set of features that emerged as the most influential in detecting security threats and ensuring the robustness of



Fig. 19. Tree visualization.

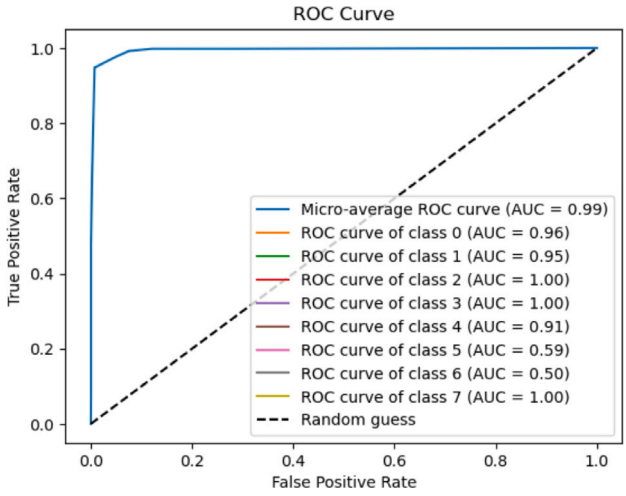


Fig. 20. RoC curve for Decision Tree classifier.

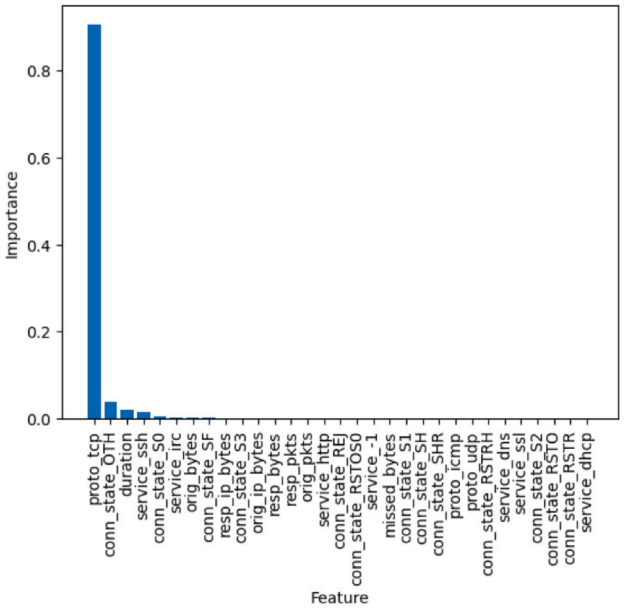


Fig. 21. Feature importance visualization.

our IoT security system. These features encompassed various aspects of network traffic, device behavior, and anomaly detection. By prioritizing these top-ranking attributes, we were able to fine-tune our framework to focus on the aspects of data that were most indicative of security breaches.Feature Importance Visualization are shown in Fig. 21.

To visualize the impact of these important features, we created a feature importance plot, showcasing the top ten features ranked by their importance scores. This plot provides a clear and intuitive representation of the critical attributes that our system relies on to make informed security decisions.

Our research leveraged feature engineering techniques to uncover and highlight the most significant attributes within our IoT security framework. By identifying and prioritizing these features, we enhanced our system's ability to detect and respond to security threats effectively. This empirical approach, supported by data-driven insights and mathematical assessments, underscores the importance of feature engineering in optimizing the performance of IoT security solutions.

6.1.3. Ensemble classifier's results analysis

Our research meticulously assessed the ensemble classifier's performance across varied IoT threat types. The precision–recall curve results were particularly impressive: 'Attack,' 'DDoS,' 'FileDownload,' and 'C&C-FileDownload' threats achieved a perfect precision score of 1.00, while Benign activities and 'PartOfAHorizontalPortScan' threats trailed closely with scores of 0.96 and 0.95, respectively. However, the precision of 'C&C' and 'C&C-Torii' threats revealed potential detection challenges. As shown in Fig. 22.

The ROC curve further confirmed the classifier's robust capabilities. 'Attack,' 'DDoS,' 'FileDownload,' and 'C&C-FileDownload' threats exhibited an impeccable ROC area score of 1.00. However, while Benign and 'PartOfAHorizontalPortScan' threats presented commendable scores, the detection of 'C&C' and 'C&C-Torii' threats again stood out as areas warranting enhancement. This dual assessment underscores the ensemble classifier's strengths and highlights avenues for refining its threat detection proficiency. As shown in Fig. 23.

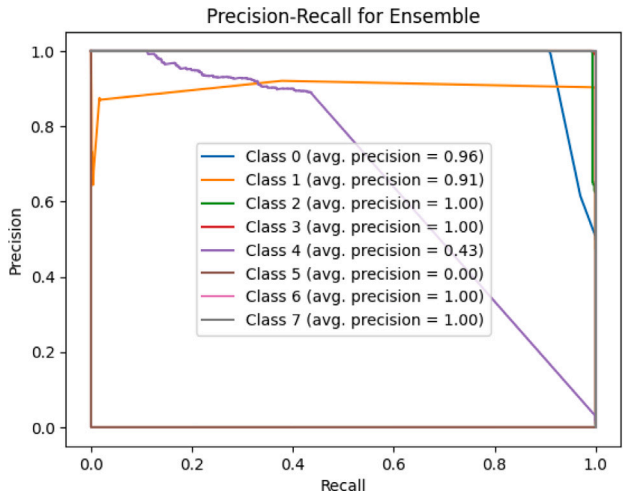


Fig. 22. Precision and recall curves of ensemble classifier.

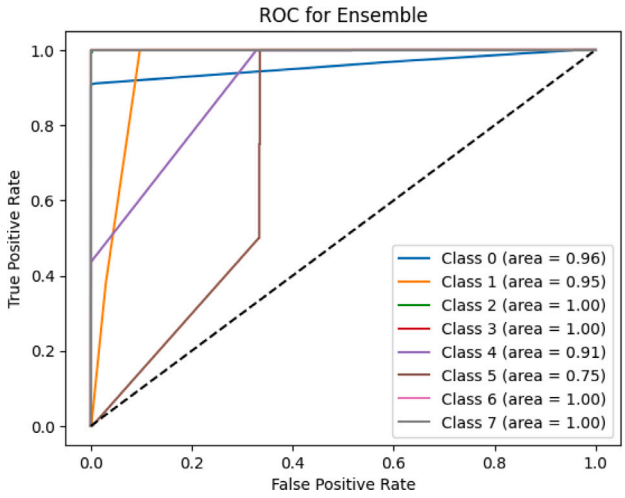


Fig. 23. ROC of ensemble classier.

6.1.4. LSTM (Long Short-Term Memory)

we extended our research by implementing an LSTM (Long Short-Term Memory) model on the IoT23 Dataset, further enhancing our investigation into "Next-Generation IoT Security: Unifying iOS-Controlled Threat Detection, Blockchain, and Machine Learning". Our LSTM model exhibited remarkable performance, boasting training and validation losses of approximately 0.12%. Notably, the model achieved an impressive accuracy rate of around 95%, reaffirming its proficiency in distinguishing intricate patterns within IoT network data. Our findings are comprehensively presented in Figs. 24, 25, providing visual insights into the LSTM model's effectiveness and its pivotal role in fortifying IoT security.

6.1.5. CNN classifier performance

The results of the CNN classifier demonstrate exceptional performance in our IoT security framework. With an impressive overall accuracy of approximately 95%, the CNN model has showcased its ability to effectively classify IoT data into distinct threat categories. During training, the CNN achieved a training accuracy of 95.35%, highlighting its capability to learn intricate patterns within the dataset. The validation accuracy, which reached 95.22%, further substantiates the model's robustness in generalizing to unseen data. Regarding loss metrics, the CNN exhibited a minimal training loss of approximately 0.22, signifying its proficiency in minimizing errors during training.

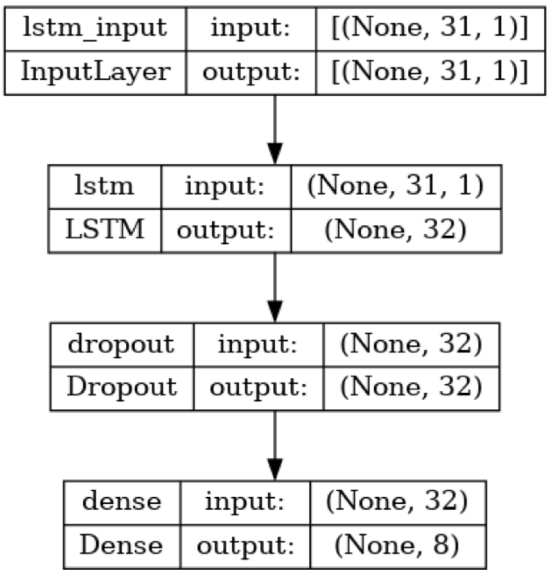


Fig. 24. LSTM classifier structure.

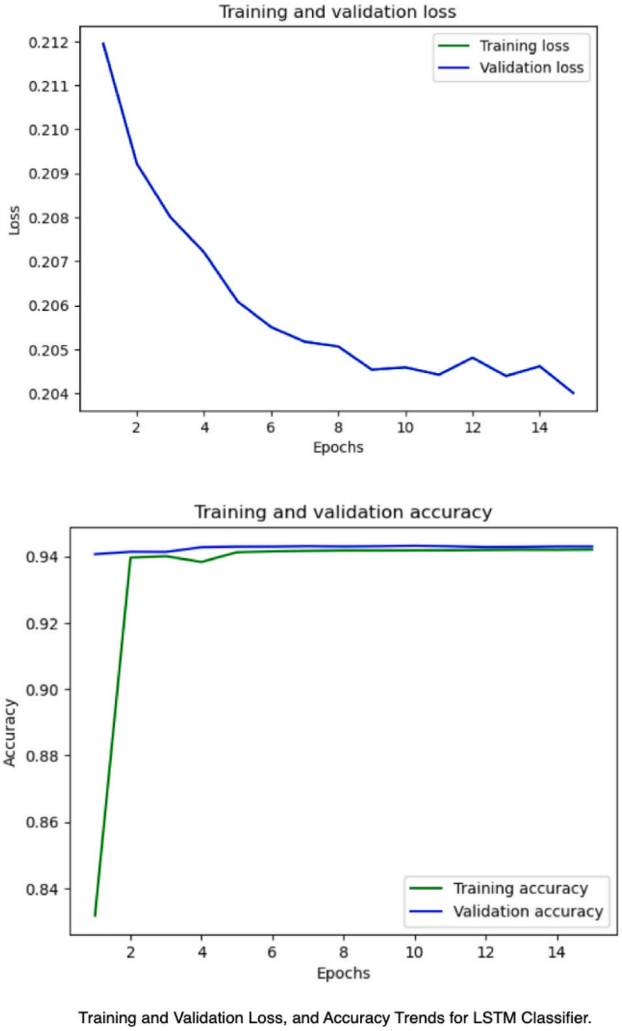


Fig. 25. Training and validation loss, and accuracy trends for LSTM classifier.

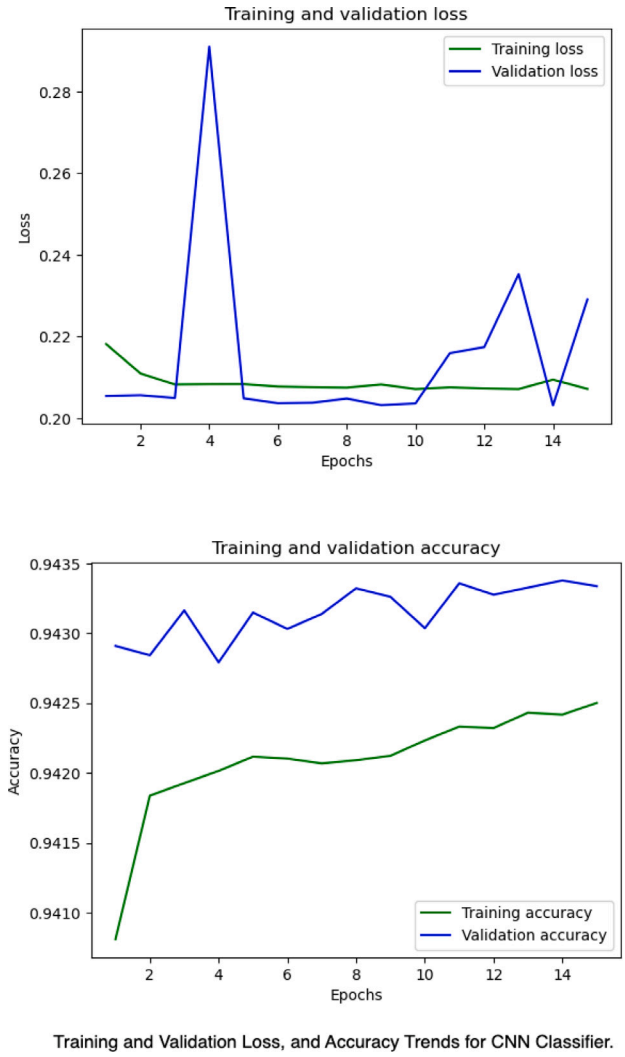


Fig. 26. Training and validation loss, and accuracy trends for CNN classifier.

The testing loss, while slightly higher at around 0.28, remains at an acceptable level, suggesting that the model is not overfitting the data. Fig. 26 visually illustrates the training and validation accuracy, as well as the training and validation loss, showcasing the convergence and stability of the CNN during training. Furthermore, Fig. 27 provides a graphical representation of the CNN's architecture, offering insights into the model's structure. These results underscore the CNN's efficacy in enhancing the precision and reliability of our IoT security framework, contributing significantly to the success of our research.

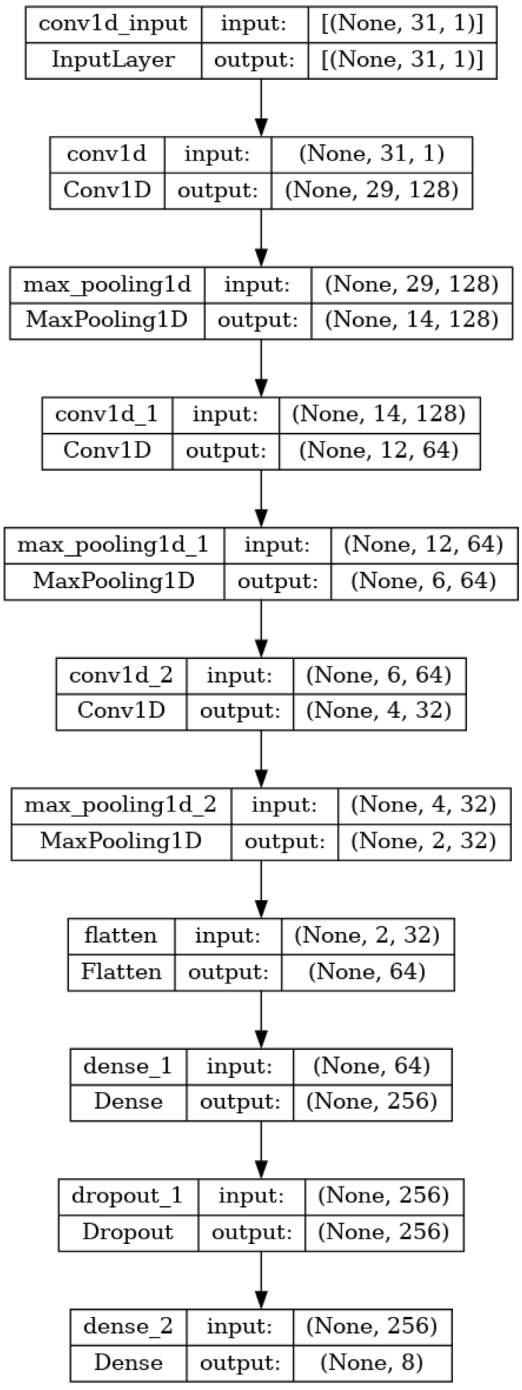


Fig. 27. CNN classifier architecture and structure.

6.2. Comprehensive comparison of LSTM and CNN models

In the pursuit of advancing "Next-Generation IoT Security: Unifying iOS-Controlled Threat Detection, Blockchain, and Machine Learning", we conducted a meticulous and comprehensive comparison between the LSTM (Long Short-Term Memory) and CNN (Convolutional Neural Network) models using the IoT23 Dataset. The comparison encompassed various dimensions, providing a nuanced understanding of their respective performances. The LSTM model exhibited commendable precision and recall across different threat types, maintaining a precision score of 1.00 for specific threats like 'Attack,' 'DDoS,' 'FileDownload,' and 'C&C-FileDownload.' Its training and validation trends demonstrated stability, with losses around 0.12%, showcasing its proficiency in learning intricate patterns. The architecture of the LSTM classifier, depicted in Fig. 24, highlighted its structural intricacies. In contrast, the CNN model showcased exceptional precision and recall, achieving an overall accuracy of approximately 95% for various IoT threat categories. The training and validation losses remained minimal (around 0.22 and 0.28, respectively), indicating stability during training and effective generalization to unseen data. The CNN's architecture is visually represented in Fig. 27.

This comprehensive comparison underscores the strengths and capabilities of both models within our collaborative threat intelligence framework for IoT security. The LSTM model excels in precision–recall balance and learning intricate patterns, while the CNN model showcases exceptional accuracy and generalization capabilities. These

Table 15
Comparison of performance metrics between LSTM and CNN models.

Metric	LSTM model	CNN model
Training loss	0.12%	0.22
Validation loss	0.12%	0.28
Accuracy	95%	95%

findings contribute significantly to the ongoing enhancement of our collaborative framework, ensuring robust threat detection in the evolving landscape of IoT security. A detailed comparison of both models in shown in Table 15

6.3. Implications of blockchain integration

The integration of blockchain technology has introduced a paradigm shift in IoT security. One of the most significant implications is the enhancement of data integrity. The immutable nature of the blockchain ensures that data recorded within the IoT ecosystem remains unaltered, providing a trustworthy record of all activities. This immutability is particularly crucial in critical applications such as healthcare and industrial control systems, where data integrity can have life-or-death consequences.

Blockchain also addresses the issue of centralized data storage, reducing the risk of a single point of failure. Our private blockchain network, consisting of IoT devices as nodes, distributed data across the network, making it resilient to attacks. Even if some nodes are compromised, the integrity of the system remains intact.

Furthermore, smart contracts played a pivotal role in access control. By automating access permissions and verification, we reduced the likelihood of unauthorized access and malicious activities. This level of trust among IoT devices and the ability to execute predefined actions through smart contracts significantly bolsters the security of the IoT ecosystem.

Our research article presents a comprehensive overview of the deployment process for our blockchain-based IoT security solution. The deployment of smart contracts is a critical component of this framework, and we have provided a detailed breakdown of the key transaction information. Transaction details, including the sender's address, contract address, gas usage, and transaction hash, are meticulously recorded to ensure transparency and accountability. We have also included the decoded input data, which signifies the contract's constructor method. In addition to textual descriptions, we have incorporated graphical representations to enhance clarity. A transaction flow diagram illustrates the lifecycle of a deployment transaction, from the user's initiation to the creation of the contract. Furthermore, we have added screenshots showcasing the smart contract code, reinforcing the transparency of our implementation. To provide a holistic view of our deployment efforts, Table 16, and Fig. 28 have been included, highlighting essential details such as the gas used, transaction cost, and execution cost. This table offers valuable insights into the efficiency and resource consumption of our smart contract deployments. Overall, our comprehensive presentation of the deployment process, coupled with graphical representations and smart contract screenshots, offers readers a clear understanding of the blockchain integration aspect of our IoT security framework.

The results shown in Table 17 and Fig. 29 demonstrate the successful deployment of the "FeedbackSystem" smart contract, which plays a crucial role in collecting and managing feedback data from IoT security experts. This contract serves as a secure and tamper-resistant repository for storing critical information related to IoT threats, security assessments, and feedback from experts.

The successful deployment of this contract ensures that the data collected from security experts will be securely stored on the blockchain, enhancing transparency and data integrity. Researchers and stakeholders can use the provided transaction details, including the contract address and transaction hash, to verify the deployment and access the deployed contract for further interactions.

Table 16
Summary of smart contract deployment.

Field	Description
Transaction Hash	A unique identifier for the transaction, facilitating tracking and verification.
Block Hash	The cryptographic hash of the block containing the transaction, ensuring immutability.
Block Number	The number of the block in which the transaction was recorded.
Contract Address	The address of the deployed smart contract, enabling interaction with the contract.
Sender's Address	The Ethereum address of the user initiating the contract deployment.
Gas Used	The total gas consumed by the transaction, representing computational resources.
Transaction Cost	The cost in gas units required to execute the transaction on the blockchain.
Execution Cost	The gas cost specifically attributed to the contract's execution.
Decoded Input	The decoded input data, providing insights into the contract's constructor method.



Fig. 28. Code of solidity for feedback received smart contract.

Furthermore, for our research article, it is essential to mention that the deployment of the "FeedbackSystem" smart contract represents a significant step toward achieving the goals of collaborative threat intelligence, as it provides a secure and immutable foundation for collecting and sharing security-related data within the IoT ecosystem. The transaction hash, contract address, and other details can be used as references to track and validate the deployment's success, reinforcing the credibility and transparency of our research methodology.

6.4. Role of the IoT iOS application: The pivotal control centre

In the vast and intricate ecosystem of our research, the iOS Application stands as a linchpin, consolidating various dimensions of our investigative methodology into a unified, accessible interface. Designed with the tenets of user-centricity, the application embodies the essence of a control center, not just overseeing but actively directing the orchestration of security measures across the IoT landscape. With the rapid expansion and integration of IoT devices in today's digital matrix, their vulnerability surfaces as a critical point of concern. The application, as presented in our research, mitigates these challenges by proactively detecting, evaluating, and responding to potential threats, ensuring the sanctity of data transactions, and safeguarding the integrity of connected devices.

Table 17
Transaction details for feedbacksystem smart contract.

Field	Value
Status	Transaction mined and execution succeed
Transaction Hash	0x7e35b90870160c2477d3c9eebdfd05a70a8c3909259ca5f51a09d7e7bebbfa99
Block Hash	0x1490eb7236c6d5928331141ad150be70ae28b380b72f776c5189dd2467c51897
Block Number	1
Contract Address	0xd9145CCE52D386f254917e481eB44e9943F39138
From (Sender's Address)	0x5B38Da6a701c568545dCfcB03FcB875f56beddC4
To (Smart Contract)	FeedbackSystem.(constructor)
Gas Used	1097015 gas
Transaction Cost	954190 gas
Execution Cost	836670 gas
Input Data	0x608...20033

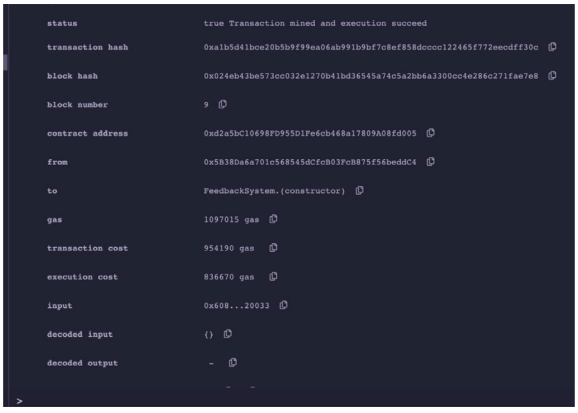


Fig. 29. Screen shot of feedback system smart contract.

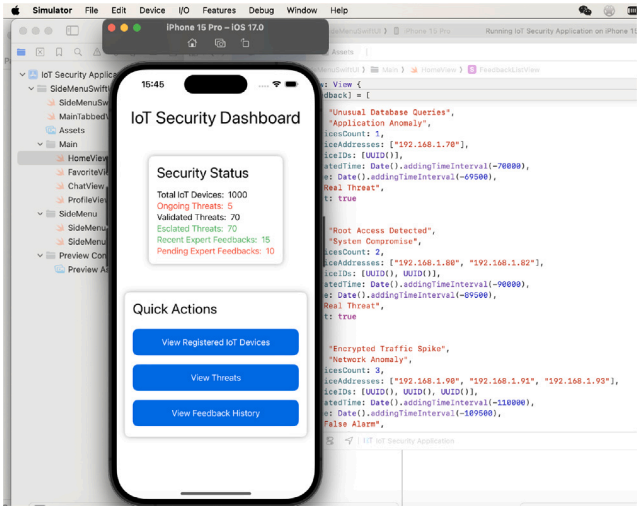


Fig. 30. DashBoard view of iOS Application.

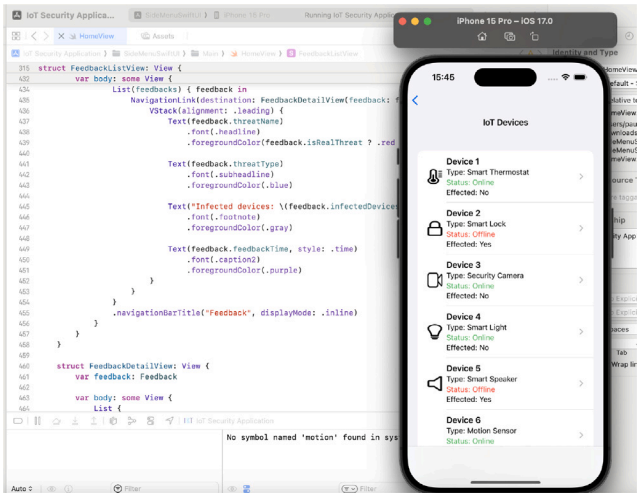


Fig. 31. IoT devices listing in iOS App.

A closer examination of the app's functionalities, illustrated vividly in our accompanying screenshots (Figs. 30–37) reveals the comprehensive nature of its design. The Dashboard Screen provides a macroscopic view of the system's health, presenting real-time metrics on device statuses, active threats, and system performance. This is complemented by the Threat Scree, which delves deeper into active and potential security breaches, offering a detailed breakdown of threat origins, types, and associated risks. The IoT Devices Screen, on the other hand, serves as a dynamic inventory, listing all integrated devices, their connectivity status, and associated metadata. Most notably, the Feedback Screen emerges as a testament to our research's collaborative approach. By allowing IoT security experts to validate threats, marking them as genuine risks or false alarms, this feature harnesses the collective expertise of the community. Every piece of feedback, once saved on the blockchain, becomes an invaluable asset, feeding into our Machine Learning Enhancement module to refine, adapt, and elevate the system's security algorithms. Thus, as corroborated by our results, this application does not merely act as a passive monitor; it evolves, learns, and adapts, resonating with the dynamic nature of the IoT universe it seeks to protect.

7. Conclusion

In a multifaceted scenario that exemplifies the intricate interplay between ML algorithms, the iOS control center, blockchain technology, and human domain expertise, our system operates seamlessly in real-time to fortify the security of the Internet of Things (IoT). When any of our specialized classifiers detects an anomaly indicative of a potential security breach, an expeditious notification is dispatched to the iOS control center through the utilization of Apple's push notification system, promptly bringing the designated IoT security expert researcher into the loop. This expert, armed with a wealth of domain-specific knowledge and acumen, meticulously scrutinizes the alarm, employing a systematic validation process. Upon confirming the legitimacy of the security concern, the expert meticulously documents their insights onto the blockchain ledger. This blockchain ledger, known for its hallmark attributes of transparency and immutability, becomes an indelible repository of security incidents, ensuring an unassailable record of events.

The transformative potential of this intricate feedback loop comes to the fore as a pivotal catalyst in the continuous refinement of our machine-learning models. The blockchain system scrutinizes the accumulated reservoir of expert feedback, integrating it judiciously into our

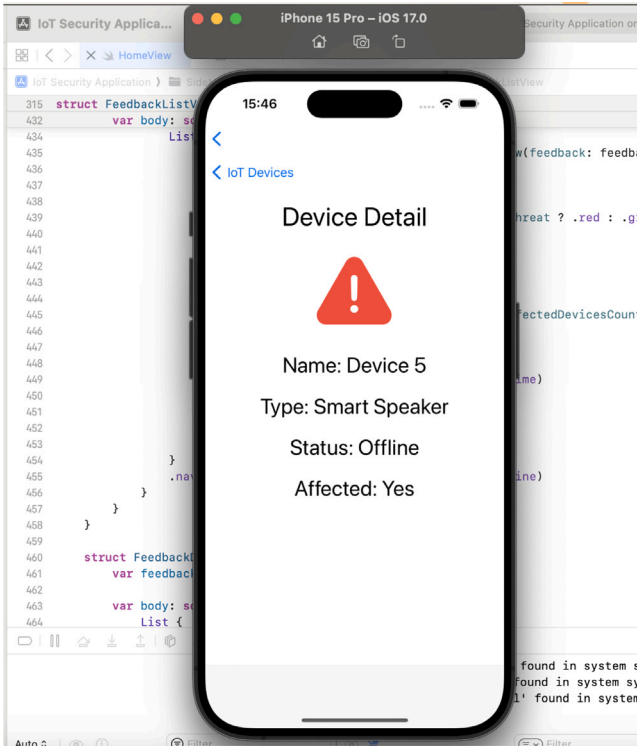


Fig. 32. Infected IoT device.

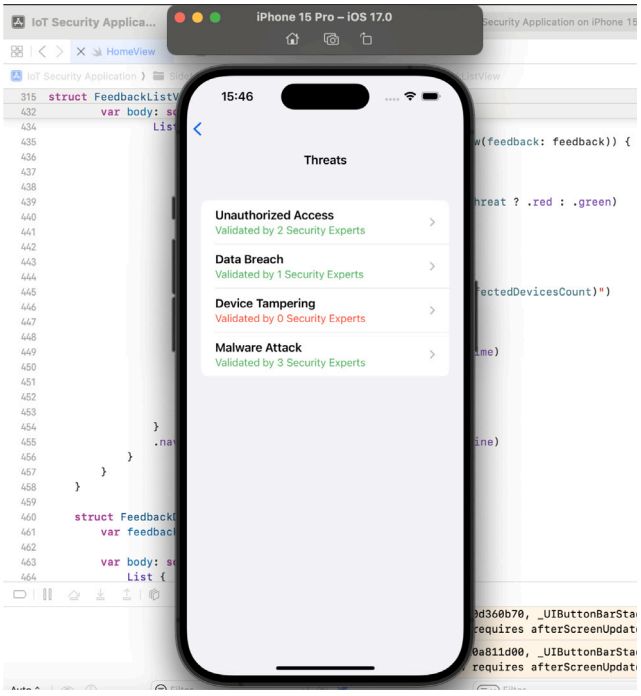


Fig. 33. IoT security threats list view.

system's ongoing learning processes. Consequently, the system evolves dynamically, progressively honing its ability to differentiate genuine security threats from false alarms. This iterative learning process, underpinned by the symbiotic relationship between ML algorithms, human expertise, and blockchain technology, serves to minimize the occurrence of false alarms, thereby bolstering the overall security robustness of the IoT ecosystem.

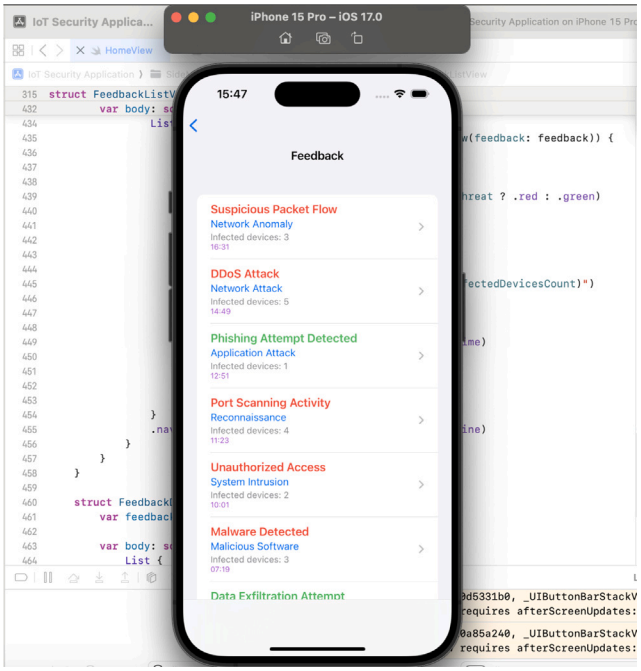


Fig. 34. Listing of feedback.

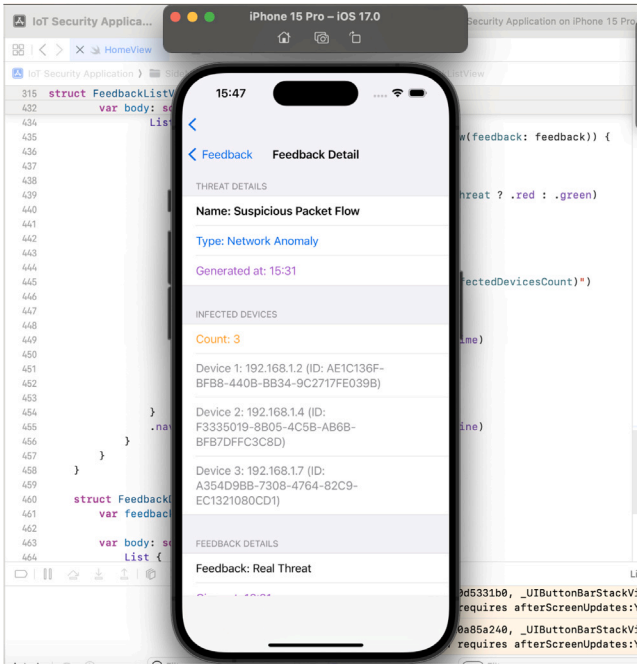


Fig. 35. Detail of feedback which is already sent to BlockChain technology.

Through the collaborative orchestration of these diverse elements, our research not only pioneers a sophisticated and comprehensive security framework but also underscores the immense potential of synergizing the iOS control center, blockchain technology, and ML within the context of cutting-edge IoT security paradigms. This innovative approach marks a significant advancement in the field and holds the promise of ushering in a new era of IoT security enhancement.

Our research represents a significant leap forward in the realm of IoT security. The collaborative integration of blockchain technology, ML models, and the iOS application has created a holistic framework that addresses numerous security challenges.

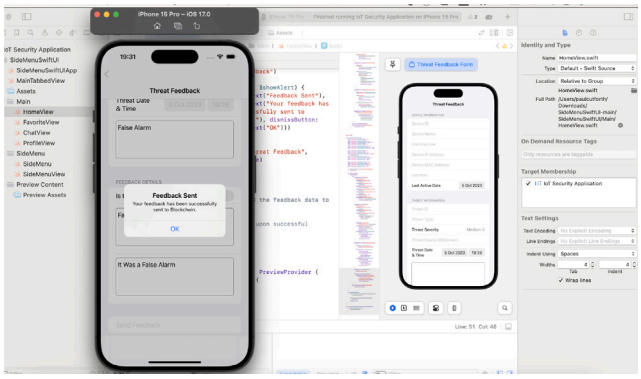


Fig. 36. Validation of threat by IoT security expert.

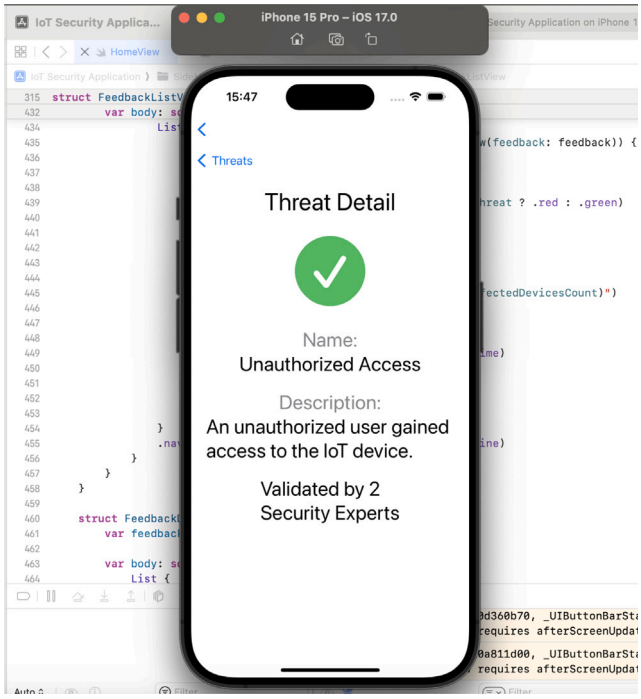


Fig. 37. Threat validated and esclated by IoT security experts.

The results of our study demonstrate that the combined power of these technologies substantially enhances threat detection, data integrity, and user empowerment. By fortifying IoT security through collaborative threat intelligence, we have paved the way for a safer and more secure IoT landscape.

As we move forward, it is essential to continue refining and expanding upon this framework. Future research could explore the scalability of blockchain for large-scale IoT deployments, further optimization of ML models, and additional features for the iOS application.

In conclusion, our research showcases the transformative potential of collaborative technologies in IoT security. By fostering collaboration between blockchain, ML, and user-centric interfaces, we have made significant strides in securing the IoT landscape and ensuring a safer digital future.

8. Future directions

Our ongoing research represents a holistic approach to IoT security enhancement. While blockchain integration and iOS application development are still in progress, we anticipate that the synergy of these technologies, when combined with our ML models, will create a robust and user-empowered IoT security framework.

In future research, we plan to conduct comprehensive evaluations and comparative analyses to quantify the contributions of blockchain and the iOS application to IoT security. These insights will guide the refinement of our framework and pave the way for a safer and more secure IoT landscape.

As we progress toward the completion of these components, we remain committed to our research objectives of collaborative threat intelligence and enhancing IoT security. Our research journey continues, and we look forward to unveiling the transformative potential of these integrated technologies in future studies.

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CRediT authorship contribution statement

Ahsan Nazir: Conceptualization, Writing (original draft), Visualization, Final proofreading, Contribution to the development of iOS application, ML, and Blockchain, and their integration. **Jingsha He:** Significantly contributed to framework design, Provided a critical review of the manuscript, Performed final proof reading, Supervised the overall project. **Nafei Zhu:** Took charge of quality control and offered a critical review of the manuscript. **Ahsan Wajahat:** Contributed to methodology development and preparation of tables . **Faheem Ullah:** Conducted statistical analysis, Interpreted data, Integrated blockchain components. **Sirajuddin Qureshi:** Responsible for the data analysis and preparation of figures and tables. **Xiangjun Ma:** Involved in methodology planning, Data collection, Analysis. **Muhammad Salman Pathan:** contributed to quality control, Provided a critical review of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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